

Mean-Field Peeling

A Complete Compact Research Report

Explicit Gaussian Normal Forms, Backward Kernels, and Local Feature-Learning Jets

Amir Joudaki's mean-field peeling research program

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Purpose. This report consolidates the original proposal, the maintained mathematical interpretation, the audited backward-kernel execution, the one-step and two-step μP calculations, and the current theorem and literature positioning. This compact edition changes layout only: it retains the complete mathematical source material while removing decorative divider pages and loose vertical spacing.

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Preface: what is established and what remains open

Mean-field derivative peeling is a proposed layerwise elimination calculus for fixed-order scalar derivative observables of wide neural networks. Its desired output is a *Gaussian normal form*: a finite directed acyclic graph whose leaves are fully specified Gaussian expectations and whose remaining operations are deterministic algebra.

Current status. The conditional Gaussian identities, finite-width chain-rule expansions, backward-kernel expectation calculation, μ P scaling, one-step tangent factorization, exact two-Euler-step Taylor factors, and deep-linear audits are explicitly derived. The general normal-form theorem, a complete concentration theorem, depth-uniform finite-state closure, and a fully enumerated nonlinear two-step channel registry remain proof obligations.

The report therefore separates exact finite-width identities, mean-field results under explicit assumptions, formal recursions, audited restricted calculations, target theorems, and open implications.

Part I. A Guided Synthesis

1. The central problem

Fix a batch of B input-label pairs and a fully connected network of fixed depth L and width n :

$$h^0(a) = x_a, \quad z^\ell(a) = W^\ell h^{\ell-1}(a), \quad h^\ell(a) = \phi(z^\ell(a)).$$

The objects of interest are scalar contractions built from outputs, losses, hidden Grams, and a fixed number of parameter or training derivatives. A representative family is

$$[\eta^r] \mathbb{E} \left[\frac{1}{n} \langle h_a^{\ell(s)}(\eta), h_b^{\ell(s)}(\eta) \rangle \right],$$

where derivative order r , step count s , batch size, depth, and observable syntax are fixed before $n \rightarrow \infty$.

The proposed calculation is:

1. expand every derivative into products of explicit weights, preactivations, and scalar activation derivatives;
2. expose every neuron and batch index;
3. select the highest active layer;
4. condition on lower layers and integrate the current Gaussian matrix;
5. execute every Stein/Onsager correction;
6. enumerate index-equality patterns and count their global width powers;
7. replace surviving row averages by deterministic Gaussian moments only after every index using them is exposed; and
8. pass the resulting boundary state to the next lower layer.

After the last peel, no random weight or width index should remain.

2. The exact local calculus

2.1 Conditional Gaussianity

Global joint Gaussianity of all deep variables is false. The reusable exact fact is layerwise conditional Gaussianity. Let

$$w \sim N(0, I_n), \quad Z_a = \frac{1}{\sqrt{n}} \sum_j w_j h_j(a).$$

Conditional on the lower-layer values $h_j(a)$, (w, Z) is jointly Gaussian and

$$\text{Cov}(w_p, Z_a \mid h) = \frac{h_p(a)}{\sqrt{n}}.$$

2.2 One- and two-weight identities

For smooth $F : \mathbb{R}^B \rightarrow \mathbb{R}$,

$$\mathbb{E}[w_p F(Z) \mid h] = \frac{1}{\sqrt{n}} \sum_a h_p(a) \mathbb{E}[\partial_a F(Z) \mid h],$$

and

$$\mathbb{E}[w_p w_q F(Z) \mid h] = \delta_{pq} \mathbb{E}[F(Z) \mid h] + \frac{1}{n} \sum_{a,b} h_p(a) h_q(b) \mathbb{E}[\partial_a \partial_b F(Z) \mid h].$$

The first branch is Wick pairing. The second is the complete Stein/Onsager correction. It cannot be deleted before global width counting.

2.3 Index ownership and mean-field replacement

An index may be summed during group m only when it appears in no lower group. A factor created by Stein differentiation belongs to the group of its preactivation source and becomes boundary data for the next peel.

The exact conditional law initially uses empirical covariances such as

$$G_{n,ab}^{\ell-1} = \frac{1}{n} \sum_j h_j^{\ell-1}(a) h_j^{\ell-1}(b).$$

One first performs conditional integration by parts, retains all lower-group factors, exposes equality patterns, and only then replaces $G_n^{\ell-1}$ by its deterministic limit. Distinguished rows require leave-one-out or leave-two-out covariances and uniform integrability.

3. The theorem program

3.1 Conservative target

An admissible observable must be defined syntactically from outputs, losses, preactivations, activations, hidden Grams, fixed-order parameter derivatives, finite products, sums, and complete tensor contractions, with every normalization declared.

Target Mean-Field Derivative Peeling Theorem. For every admissible scalar derivative observable O_n , a deterministic transformation recursively eliminates the maximal active layer. Each peel emits explicit Gaussian-integral coefficients multiplying admissible lower-group states. Under smoothness, moment, leave-one-out, conditional-CLT, concentration, and uniform-integrability hypotheses,

$$\lim_{n \rightarrow \infty} \mathbb{E}[O_n] = \text{Eval}(\text{Peel}(O_n)).$$

Every covariance, integrand, activation derivative, equality pattern, and coefficient in the output is mechanically specified.

This is a target theorem, not a result proved by the current examples.

3.2 Finite-state strengthening

The strongest useful extension would prove that, at fixed batch size, derivative order, step count, and observable schema, the boundary signatures close on finitely many depth-independent state types. After sharing common states, the normal-form DAG would then have at most

$$C_{B,r,O}L$$

nodes. The present nonlinear registry is not fully enumerated and its channel count may depend on depth. Thus $O(L)$ complexity remains an open target.

3.3 Feature-learning jets

For s training steps and Taylor order p , the coefficients

$$[\eta^k] \mathbb{E}[\mathcal{L}_a^{(s)}(\eta)], \quad [\eta^k] \mathbb{E}[G_{ab}^{\ell,(s)}(\eta)], \quad k \leq p,$$

form an order- p *feature-learning jet*. A peeling theorem would give each admissible coefficient an explicit Gaussian normal form.

A fixed jet does not imply convergence of the Taylor series, control at a fixed nonzero learning rate, or a complete positive-time training trajectory.

4. The audited μP laboratory

4.1 Network and kernels

The worked model has three hidden layers:

$$\begin{aligned} z_i^1(a) &= \frac{1}{\sqrt{d_0}} \sum_j W_{ij}^1 x_{aj}, & z_p^2(a) &= \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 h_i^1(a), \\ z_r^3(a) &= \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 h_p^2(a), & f_a &= \frac{1}{n} \sum_r v_r h_r^3(a), \end{aligned}$$

with iid standard-Gaussian raw parameters and

$$\mathcal{J} = \frac{1}{B} \sum_c (f_c - y_c)^2, \quad \theta^+ = \theta - n\eta \nabla_{\theta} \mathcal{J}.$$

Let

$$\begin{aligned} G_{ab}^0 &= \frac{x_a^\top x_b}{d_0}, & G_{ab}^\ell &= \mathbb{E}[\phi(Z_a^\ell) \phi(Z_b^\ell)], \\ D_{ab}^\ell &= \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell)], & Z^\ell &\sim N(0, G^{\ell-1}). \end{aligned}$$

4.2 Backward sensitivity

Define

$$\delta_i^\ell(a) = n \frac{\partial f_a}{\partial z_i^\ell(a)}, \quad \Pi_{n,ab}^\ell = \frac{1}{n} \sum_i \delta_i^\ell(a) \delta_i^\ell(b).$$

The top-down Wick–Stein calculation gives the expectation limits

$$\Pi^3 = D^3, \quad \Pi^2 = D^2 \odot D^3, \quad \Pi^1 = D^1 \odot D^2 \odot D^3.$$

Random-kernel convergence requires a separate four-path concentration peel.

4.3 One training step

The optimizer-metric tangent kernel is exactly

$$K_{n,ab} = G_{n,ab}^3 + G_{n,ab}^2 \Pi_{n,ab}^3 + G_{n,ab}^1 \Pi_{n,ab}^2 + G_{ab}^0 \Pi_{n,ab}^1.$$

Under the joint concentration assumptions,

$$\begin{aligned} K_{ab} &= G_{ab}^3 + G_{ab}^2 D_{ab}^3 + G_{ab}^1 D_{ab}^2 D_{ab}^3 \\ &\quad + G_{ab}^0 D_{ab}^1 D_{ab}^2 D_{ab}^3. \end{aligned}$$

Consequently,

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[(f_a^+ - y_a)^2] = -\frac{4}{B} y_a \sum_c K_{ac} y_c.$$

The annealed hidden-Gram linear coefficient vanishes,

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[G_{n,ab}^{\ell,+}] = 0,$$

while the first generic deterministic coefficient is quadratic:

$$\lim_{n \rightarrow \infty} [\eta^2] \mathbb{E}[G_{n,ab}^{\ell,+}] = C_{ab}^\ell.$$

The detailed recursion contains a fresh off-diagonal Gaussian field; replacing the empirical random operator by its mean before squaring gives a wrong coefficient.

4.4 Two Euler steps

For $F(\theta) = -n\nabla \mathcal{J}(\theta)$,

$$\theta^{(2)} = \theta + 2\eta F + \eta^2 DF[F] + O(\eta^3).$$

For a hidden-Gram observable O , put

$$L = DO[F], \quad C = \frac{1}{2}D^2O[F, F], \quad R = DO[DF[F]].$$

Then the cumulative and second-update expansions are

$$O(\theta^{(2)}) = O(\theta) + 2\eta L + \eta^2(4C + R) + O(\eta^3),$$

$$O(\theta^{(2)}) - O(\theta^{(1)}) = \eta L + \eta^2(3C + R) + O(\eta^3).$$

The factors 4 and 3 are exact. The general nonlinear reduction of R is a formal multichannel schema pending a complete registry and convergence theorem.

4.5 Deep-linear audit

For $\phi(z) = z$, with $\kappa = 2/B$ and $\sigma = G^0 y$,

$$(C^1, C^2, C^3)_{ab} = (1, 5, 14)\kappa^2\sigma_a\sigma_b,$$

$$(R^1, R^2, R^3)_{ab} = (6, 14, 20)\kappa^2\sigma_a\sigma_b.$$

Thus the cumulative two-step coefficients are (10, 34, 76) and the second-update coefficients are (9, 29, 62), each multiplying $\kappa^2\sigma_a\sigma_b$.

5. Lessons, positioning, and reading route

5.1 Corrections forced by execution

1. Fix raw versus effective parameter conventions before counting width.
2. Peel from the highest group downward.
3. Perform exact conditional Stein integration before covariance replacement.
4. Retain Stein-created lower-group factors as boundary data.
5. Do not discard a zero-mean family before checking a second copy.
6. Separate expectation limits from concentration.
7. Do not differentiate a limiting zero as though it were identically zero.
8. At later steps, track differentiated states and every cross-covariance.
9. Treat network states as nonlinear functions of Gaussian base channels, not as one jointly Gaussian state.
10. Use a safe fixed multiple qB for Gaussian dimension; exactly B is not established generally.

5.2 Position relative to Tensor Programs

Tensor Programs already gives constructive recursive limiting semantics for broad finite neural computations, including μP feature-learning limits and matrix-reuse Onsager corrections. The defensible proposed contribution of peeling is narrower: a source-to-source normal-form compiler for contracted derivative observables that explicitly executes every derivative expansion, Stein correction, Wick contraction, equality partition, and width count.

5.3 How to use the detailed parts

Part II preserves both the original notes and the maintained mathematical program. Part III is the audited calculation-level case study. Part IV states the sharp theorem target, the Tensor Programs contrast, required nonclaims, and the decisive remaining proof obligations. Repetition across these parts is intentional: each preserves a distinct layer of the research state.

Part II. The Maintained Peeling Program

The Mean-Field Peeling Program

Status: Living research document

Canonical format: Markdown

Initial maintained version: 2026-08-01

Conceptual source: Amir Joudaki's raw notes and NTK sample execution

Current role: Preserve the original proposal, record the best current formal interpretation, and accumulate checked executions of the method.

Maintenance convention

This document has two logically different components.

1. **Source record.** Part I preserves the original program as proposed. Later mathematical corrections must not be silently written back into this section.
2. **Maintained interpretation.** Parts II onward record the current best understanding. These sections may be revised as notation, hypotheses, state spaces, and proofs improve.

Every substantive future revision should add an entry to the change log. Statements are labeled as definitions, exact identities, programmatic claims, derived examples, or open proof obligations. A programmatic claim is not yet a theorem.

Document map

- Part I¹ preserves the original proposal.
- Part II² gives the current formal interpretation and reusable algorithm.

¹<#part-i-original-source-notes>

²<#part-ii-current-maintained-understanding>

- Part III³ explains how initialization peeling interfaces with gradient-flow feature learning.
- Part IV⁴ performs the first detailed execution.
- Part V⁵ isolates small Wick, Stein, and equality-pattern examples.
- Part VI⁶ records proof obligations and next executions.
- Appendix A⁷ preserves the supplied NTK sample's mathematical content.

The companion `MUPTRAININGPEELINGCASESTUDY.md`⁸ consolidates the complete backward-kernel execution with the one-step and two-step μP training calculations, retaining the low-level scalarization, Stein corrections, equality-pattern counts, and deep-linear audits.

Part I. Original source notes

This part preserves the original raw proposal in lightly normalized Markdown. Spelling and display formatting have been cleaned where necessary, but the conceptual content has not been changed. The original LaTeX NTK sample is summarized here and preserved as a source snapshot in Appendix A.

1. Baseline setup

Use a canonical multilayer perceptron with preactivation

$$z^\ell = W^\ell h^{\ell-1}, \quad h^\ell = \phi(z^\ell), \quad h^0 = x, \quad f(x) = z^L.$$

For a single sample (x, y) , use the loss

$$\mathcal{L}(x, y) = (f(x) - y)^2.$$

Initialization and scaling are intended to follow the μP mean-field feature-learning setup. For convenience, all hidden layers may have width n , and the mean-field limit is $n \rightarrow \infty$.

2. Group ℓ

Suppose z^ℓ is the preactivation generated by the linear weights W^ℓ , with activation ϕ applied to it. Any object that is a W^ℓ term, a z^ℓ term, a $\phi(z^\ell)$ term, a higher activation derivative, or another scalar function of z^ℓ , belongs to group ℓ .

3. Explicit terms

The terms above involve direct weights, preactivations, and postactivations with scalar activation functions such as ϕ , ϕ' , and ϕ'' . These are called explicit group- ℓ terms.

An object such as the backward error

$$\delta^\ell = \frac{\partial \mathcal{L}}{\partial z^\ell}$$

is implicit rather than explicit. Chain rules, product rules, and higher-order derivative rules can expand any finite-order derivative object into a sum of products of explicit terms. The individual products are called product terms.

4. Scalarization

Product terms are normally matrix or higher-order tensor contractions and are therefore noncommutative before their indices are exposed. Any tensor contraction can be written in scalar form by introducing enough indices. For example,

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

After doing this across a product term, every matrix or tensor product becomes an ordinary scalar product together with explicit index sums. The resulting expressions are scalarized product terms. They involve only scalar weight entries, scalar preactivation entries, and scalar functions of preactivations.

Because the goal is an initialization expectation, linearity allows the expectation to be applied term by term.

5. Grouping and reordering from bottom to top

Within a scalarized product term, scalar multiplication is commutative. Terms can therefore be grouped by layer and ordered from the lowest group index to the highest, with the highest group placed at the far right.

6. Pushing the top-group expectation forward

Suppose the highest active group is m . The intended step is to integrate the fresh randomness of W^m only after all lower-group factors have been placed outside that expectation. Informally, lower-layer randomness is separate from the fresh randomness in W^m .

7. Mean-field multivariate Gaussian limit

The forward pass produces layerwise Gram matrices transported by Gaussian expectations. Schematically,

$$G^{\ell+1} = \mathbb{E}[\phi(X)\phi(Y)^\top], \quad (X, Y) \sim N(0, G^\ell).$$

In the mean-field limit, each neuron row of preactivations is described by a deterministic covariance matrix. Distinct rows are independent. Weight elements are independent unless their indices agree. Weights and same-row preactivations possess a small covariance controlled by the width scaling.

The intended result is that the relevant top-layer weight and preactivation coordinates can be treated through a prescribed joint Gaussian structure.

³<#part-iii-feature-learning-interface>

⁴<#part-iv-worked-execution-the-backward-sensitivity-kernel>

⁵<#part-v-small-reusable-examples>

⁶<#part-vi-open-proof-and-development-program>

⁷<#appendix-a-original-ntk-sample-execution>

⁸<MUP_TRAINING_PEELING_CASE_STUDY.md>

8. Wick/Isserlis contractions

If a product contains only linear Gaussian terms, Wick's or Isserlis' theorem converts the expectation of the product into a sum over pairings.

9. Stein/Onsager corrections

If nonlinear functions of preactivations occur, Stein's lemma is used to separate the weight–preactivation dependence. Informally, this produces a pseudo-independent contribution plus correction terms involving derivatives of the nonlinear functions and fewer explicit weight factors.

One product term may therefore become a sum of simpler product terms.

10. Weight–activation pseudo-independence terms

Applying the Stein/Onsager step inductively to the highest group is intended to produce terms in which top-layer weights can be integrated separately from top-layer activation-related factors, together with explicit correction branches.

11. Wick contractions on the remaining weight elements

Once the weight dependence has been isolated, Wick/Isserlis contractions remove the remaining weights. Some contraction patterns vanish and others survive, producing Kronecker or Dirac delta constraints among neuron indices.

12. Mean-field peeling

After the top-layer weights have been removed, the remaining nonconstant top-group objects are functions of top-layer preactivations. The original scalarization contains large index sums. Wick pairings break these sums into a small number of equality patterns.

For example,

$$\frac{1}{n^2} \sum_{i,j} \phi(z_i) \phi(z_j) = \left(\frac{1}{n} \sum_i \phi(z_i) \right)^2 \longrightarrow (\mathbb{E} \phi(Z))^2,$$

whereas an enforced equality pattern gives

$$\frac{1}{n} \sum_i \phi(z_i)^2 \longrightarrow \mathbb{E}[\phi(Z)^2].$$

Each equality pattern defines a family of sums with its own mean-field limit and width scaling. After evaluating these sums, the entire effect of the top group is replaced by scalar Gaussian expectations. All remaining stochastic terms involve lower groups only.

Thus an expectation of scalarized products with highest group m is reduced to a weighted sum of expectations of scalarized products with highest group at most $m - 1$. This is the mean-field peeling step.

13. Repeated peeling

Repeat the same operation one group at a time. Eventually every term is reduced to products of explicit Gaussian expectations of the form

$$\mathbb{E} [f_1(Z_1) f_2(Z_2) \dots f_r(Z_r)],$$

where the covariance of the preactivations is determined by the layerwise forward Gram matrix and the functions are identity, ϕ , ϕ' , ϕ'' , or higher derivatives.

For a fixed batch of size B , the primitive Gaussian integration is intended to be finite-dimensional and controlled by the fixed batch covariance matrices.

14. Practical tips from the original proposal

14.1 Retain only leading width patterns

Only leading surviving width patterns matter for a leading-order mean-field result. A pattern with fewer free assignments may be neglected if it cannot recover the lost width order in later lower-layer summations.

The phrase “contribute similarly downstream” is important: local top-layer counts alone are insufficient if two patterns create different lower-layer sums.

14.2 Recursive scalarization

Full scalarization of every layer at once may be prohibitively complex. Instead, expose and scalarize only the current top group. After it is peeled, repeat on the next group.

14.3 Recursive states and dynamic programming

The equality patterns and lower-layer boundary signatures created by one peel may close on a finite family of states. If so, the computation can be written as a recursive or dynamic-programming system.

15. Original headline claim

The original intended headline was:

Any object involving derivatives or higher-order derivatives at initialization of a depth- L MLP in the mean-field regime, with arbitrary but constant $O(1)$ maximum derivative order and a fixed batch of B samples, can be computed through a set of $O(L)$ recursive equations involving only explicit Gaussian expectations of layerwise preactivations passed through identity, ϕ , ϕ' , ϕ'' , and higher derivatives.

16. Original feature-learning motivation

Under a μP mean-field feature-learning parameterization, the change in a hidden-layer Gram matrix after one step can be expanded to fixed order in the learning rate using higher derivatives of the network at initialization. At later steps, gradient corrections can similarly be expanded using higher derivatives.

In the gradient-flow scaling

$$t \rightarrow \infty, \quad \eta \rightarrow 0, \quad t\eta = \tau,$$

the coefficients become time derivatives of the training trajectory at initialization. Each such coefficient is intended to be an object handled by the peeling program.

17. Original conjecture

For a neural network evaluated on fixed inputs and possessing a well-defined mean-field learning limit, any scalar observable such as loss or a hidden-layer Gram matrix, provided its training curve is locally smooth, has initialization derivatives expressible as sums of products of explicit Gaussian integrals. Their covariance matrices are finite and determined by the input, output, architecture, initialization, and loss configuration.

18. Role of the original NTK sample

The supplied LaTeX note attempted a first-principles NTK derivation using forward covariance recursions, backward Jacobian moments, Wick contractions, and Stein corrections. Its intended lessons were:

- forward preactivation covariances obey a deterministic Gaussian recursion;
- backward influence can be propagated recursively;
- the classical NTK is built from forward activation kernels and derivative-transport factors;
- diagonal and off-diagonal backward patterns must be distinguished;
- off-diagonal corrections should be checked rather than discarded by an independence assumption.

The sample is a source of intuition, not yet an adopted proof. Its notation changes weight-index conventions, switches from square loss to a linear-output surrogate, and leaves several second-moment and off-diagonal states incompletely defined.

Part II. Current maintained understanding

19. The central mathematical operation

The current best formulation is:

Mean-field peeling is layerwise conditional Gaussian elimination of scalar contraction diagrams.

It is not based on global Gaussianity and is not based on unconditional independence between a weight matrix and the preactivations generated by that matrix.

The operation has four exact components:

1. expose the current highest-layer contraction;
2. condition on all preceding layers;
3. remove the fresh Gaussian variables using Wick and Stein identities;
4. convert top-neuron row sums into Gaussian moments while passing the remaining contraction state to the next lower layer.

20. Notation and scaling ledger

For the maintained formulation, let

$$A_{ij}^\ell = s_{\ell,n} \theta_{ij}^\ell$$

denote the effective weight used in the forward pass, where θ^ℓ is the raw parameter coordinate. Write

$$z_i^\ell(a) = \sum_j A_{ij}^\ell h_j^{\ell-1}(a) + \sigma_{b,\ell} b_i^\ell.$$

For fan-in initialization,

$$A_{ij}^\ell \sim N\left(0, \frac{\sigma_{w,\ell}^2}{n_{\ell-1}}\right).$$

A complete peeling problem must declare:

- all widths and their limiting ratios;
- the variance and multiplier of each effective weight;
- the raw parameter with respect to which derivatives are taken;
- output dimension and readout scaling;
- bias scaling;
- layerwise optimizer or μP metric factors;
- the target normalization and desired asymptotic order;
- whether the target is an expectation, a deterministic limit, or a fluctuation law.

No width power counting is meaningful before this ledger is fixed.

21. Exact layerwise conditional Gaussian law

Let

$$\mathcal{F}_{\ell-1} = \sigma(A^1, b^1, \dots, A^{\ell-1}, b^{\ell-1}).$$

Conditional on $\mathcal{F}_{\ell-1}$, the row blocks

$$(A_{i,:}^\ell, b_i^\ell, Z_i^\ell), \quad Z_i^\ell = (z_i^\ell(1), \dots, z_i^\ell(B)),$$

are independent centered jointly Gaussian vectors. Exactly at finite width,

$$Q_{n,ab}^\ell := \text{Cov}(z_i^\ell(a), z_i^\ell(b) \mid \mathcal{F}_{\ell-1}) = \frac{\sigma_{w,\ell}^2}{n_{\ell-1}} \sum_j h_j^{\ell-1}(a) h_j^{\ell-1}(b) + \sigma_{b,\ell}^2,$$

$$\text{Cov}(A_{ij}^\ell, A_{i'j'}^\ell) = \delta_{ii'} \delta_{jj'} \frac{\sigma_{w,\ell}^2}{n_{\ell-1}},$$

and

$$\text{Cov}(A_{ij}^\ell, z_{i'}^\ell(a) \mid \mathcal{F}_{\ell-1}) = \delta_{ii'} \frac{\sigma_{w,\ell}^2}{n_{\ell-1}} h_j^{\ell-1}(a).$$

This conditional statement replaces the original global joint-Gaussian statement. Unconditionally, the covariance contains random lower-layer activations, so the joint law across multiple layers is generally a Gaussian mixture rather than a single Gaussian vector.

22. Deterministic forward state

Under standard law-of-large-numbers and moment assumptions,

$$Q_n^\ell \longrightarrow Q^\ell,$$

where

$$Q_{ab}^1 = \sigma_{w,1}^2 \frac{(x^a)^\top x^b}{n_0} + \sigma_{b,1}^2,$$

and

$$Q_{ab}^{\ell+1} = \sigma_{w,\ell+1}^2 \mathbb{E}_{Z \sim N(0, Q^\ell)} [\phi_\ell(Z_a) \phi_\ell(Z_b)] + \sigma_{b,\ell+1}^2.$$

Labels do not enter these initialization covariances. They enter through loss derivatives and training-time vector fields.

23. Explicit atoms and admissible observables

An explicit group- ℓ atom is:

- an effective or raw layer- ℓ parameter coordinate;
- a bias coordinate;
- a preactivation coordinate $z_i^\ell(a)$;
- a scalar function $\psi(z_i^\ell(a))$, including activation and loss derivatives.

The initial admissible observable class is:

Width-normalized scalar contractions of finitely many parameter, input, or training-time derivatives of the network, loss, or a smooth permutation-invariant observable, evaluated on a fixed batch at initialization.

This class deliberately requires a scalar contraction. An uncontracted parameter Hessian with a width-growing number of free indices is not itself one scalar peeling state.

Ratios, inverses, maxima, spectral edges, and other global nonlinear functions require additional nondegeneracy or regularity arguments even when their ingredients can be peeled.

24. Scalar contraction normal form

After chain-rule, product-rule, and Faà di Bruno expansion, each scalar term takes the form

$$T_n = c_n \sum_{\iota} \mathbf{1}\{\mathcal{C}(\iota)\} \prod_{r=1}^{R_W} A_{i_r j_r}^{\ell_r} \prod_{s=1}^{R_Z} \psi_s(z_{u_s}^{k_s}(a_s)).$$

The state records:

- all neuron indices and their layers;
- all fixed batch labels;
- equality and inequality constraints;
- the derivative order of each scalar function;
- all explicit powers of width;
- all open indices that will be passed to lower layers.

This is the precise object on which peeling operates.

25. Exact generalized Wick–Stein identity

Let $X_1, \dots, X_r$ be explicit Gaussian parameter coordinates in the current top group and let $F(Z)$ be the product of all current top-preactivation functions. Conditional on the lower filtration,

$$\mathbb{E} \left[\prod_{q=1}^r X_q F(Z) \right] = \sum_{\pi} \left(\prod_{\{q,t\} \in \pi} \text{Cov}(X_q, X_t) \right) \sum_{\alpha: U(\pi) \rightarrow \mathcal{J}_Z} \left(\prod_{q \in U(\pi)} \text{Cov}(X_q, Z_{\alpha(q)}) \right) \mathbb{E} [\partial_{\alpha(U(\pi))} F(Z)]. \quad (25.1)$$

The outer sum is over partial matchings π of the weight occurrences, and $U(\pi)$ is the unmatched set.

- A matched pair is a weight–weight Wick contraction.
- An unmatched weight is attached to a preactivation through Stein’s identity.
- Every attachment differentiates the nonlinear top factor and inserts a lower-layer activation through the weight–preactivation covariance.

This exact finite identity is the correct replacement for the informal statement that Stein’s lemma creates an independent copy plus one correction.

The complete matchings are the pseudo-independent terms. Partial matchings are the Onsager branches. Whether Onsager branches survive is decided by global power counting, not by terminology.

26. Equality partitions and Gaussian row moments

For top-layer row indices $i_1, \dots, i_q$, decompose the sum by exact equality partition:

$$\sum_{i_1, \dots, i_q} = \sum_{\pi \in \mathcal{P}_q} \sum_{\ker(i)=\pi}.$$

A partition with $|\pi|$ distinct row blocks has

$$n_\ell^{|\pi|} (1 + O(n_\ell^{-1}))$$

assignments, subject to width ratios and existing constraints.

For one row block, define

$$\Gamma_\ell[(a_1, \psi_1), \dots, (a_s, \psi_s); Q] := \mathbb{E}_{Z \sim N(0, Q)} \left[\prod_{r=1}^s \psi_r(Z_{a_r}) \right].$$

Factors on distinct top rows become products of such moments. Thus each primitive integral is at most B -dimensional, although a diagram may contain several independent row blocks.

27. Global width power counting

The width degree of a diagram includes:

- one positive width power for each free neuron-index block;
- negative powers contributed by weight covariances;
- negative powers contributed by weight–preactivation attachments;
- the observable normalization;
- raw-to-effective parameter multipliers;
- μP learning-rate or metric factors;
- any lower-layer sums created by Stein attachments.

A pattern may be discarded only after its entire lower-layer boundary signature has been accounted for. Locally subleading terms can become comparable if they create additional lower-layer sums, while locally comparable terms can become subleading when they must later satisfy an extra equality.

For fluctuation observables, subleading diagrams may be the target and must not be discarded.

28. Replacing empirical covariance by deterministic covariance

After exact top-layer integration, coefficients initially depend on the lower-random empirical covariance Q_n^ℓ . For a uniformly integrable leading-order state,

$$Q_n^\ell \rightarrow Q^\ell \implies \Gamma_\ell[\dots; Q_n^\ell] \rightarrow \Gamma_\ell[\dots; Q^\ell].$$

For centered or fluctuation-amplified states, the zeroth-order replacement may erase the leading term. One then expands in $Q_n^\ell - Q^\ell$. Price's identity keeps the expansion within the same Gaussian derivative algebra:

$$D \mathbb{E}_{Z \sim N(0, Q)} [F(Z)] [H] = \frac{1}{2} \sum_{a,b} H_{ab} \mathbb{E} [\partial_{ab} F(Z)],$$

with the standard symmetric-covariance convention.

29. Formal recursive algorithm

Inputs

1. Architecture, widths, and activation functions.
2. Effective and raw parameter scalings.
3. Output and bias normalization.
4. Optimizer or μP metric scaling when training derivatives are involved.
5. Fixed input-label batch.
6. Scalar normalized observable.
7. Maximum derivative order.
8. Desired width order and convergence mode.

One peeling pass

1. Expand only enough derivative rules to expose the current highest group.
2. Scalarize the top-group contractions.
3. Record exact width powers and lower-layer boundary indices.
4. Reorder scalar atoms by group.
5. Let p be the highest active group.
6. Condition on $\mathcal{F}_{p-1}$.
7. Partition top-row indices into compatible equality patterns.
8. Apply the generalized Wick–Stein identity until no explicit group- p Gaussian parameter remains.
9. Expand every differentiated activation product.
10. Substitute exact conditional covariances.
11. Resolve Kronecker deltas and update the lower-layer boundary state.
12. Perform global width power counting.
13. Convert surviving top-row functions to Gaussian moments at Q_n^p .
14. Replace Q_n^p by Q^p , or retain the necessary covariance expansion.

15. Combine algebraically identical lower states.
16. Repeat at group $p - 1$.
17. At layer zero, evaluate deterministic input and label contractions.

Convergence pass

If the result is claimed to hold for a typical initialization rather than only in expectation, repeat the procedure for the square, covariance, or relevant cumulants and prove concentration.

30. Dynamic-programming form

A recursive state stores only the active open contraction legs, derivative orders, batch labels, equality partitions, lower-layer constraints, and accumulated width degree.

For a finite state family $\mathcal{S}$, one peel has the form

$$V_p(s) = \sum_{s' \in \mathcal{S}} T_p(s, s'; Q^p) V_{p-1}(s'),$$

where each transition entry is a finite sum of scaling constants and Gaussian moments.

Thus

$$V_L = T_L(Q^L) \dots T_1(Q^1) V_0.$$

The number of layer transitions is $O(L)$. The state dimension can depend combinatorially on derivative order, batch size, observable type, and requested asymptotic order. Closure on a state space independent of depth is a proof obligation, not an automatic consequence of fixed derivative order.

31. Current programmatic claim

The strongest claim currently justified as a program specification, rather than a theorem, is:

Fix a batch size B , derivative order R , depth L , and a Gaussian fully connected architecture whose hidden widths tend to infinity with fixed positive ratios. Let O_n be an $O(1)$ -normalized admissible scalar contraction of the initialization jet of order at most R . Assume sufficient activation smoothness and moment bounds, deterministic convergence of the forward empirical Gram matrices, uniform integrability of all retained states, and finite-state closure of the layerwise contraction rules. Then the leading large-width expectation of O_n is computable through L layer-local peeling transitions. Every transition coefficient is a finite linear combination of explicit batch-dimensional Gaussian expectations of scalar activation functions and their derivatives, evaluated at deterministic layerwise preactivation covariances. If second-moment peeling proves vanishing variance, the same expression is the limit in probability.

This remains a theorem schema until the admissible class, required regularity, state closure, normalization classes, and convergence arguments are proved.

32. Important current qualifications

1. The width limit presently assumes fixed depth, batch size, and derivative order. Depth growing with width requires new uniform estimates.
2. “Any scalar observable with a mean-field limit” is too broad without an admissibility condition.
3. Expectation, convergence in probability, and convergence in distribution are distinct outputs.
4. A scalar output layer has no neuron-row law-of-large-numbers factor and must be handled separately.
5. ReLU requires smoothing, weak derivatives, or distributional integration by parts.
6. Bias-preactivation covariances can be $O(1)$; bias Stein branches are not automatically negligible.
7. The activation derivative order required by peeling may exceed the nominal derivative order of the original observable.
8. A noncentered readout can require backward first-moment and off-diagonal states.
9. A zero-initialized readout can make the initialization backward kernel vanish, moving the first nontrivial state to a training-time derivative.

Part III. Feature-learning interface

33. Gradient-flow derivatives

Let the scaled gradient flow be

$$\dot{\theta}_n(\tau) = -P_n \nabla_{\theta} \mathcal{R}_n(\theta_n(\tau)),$$

where P_n is the layerwise optimizer metric or μP preconditioner. Define the differential operator

$$\mathcal{V}_n := -(P_n \nabla \mathcal{R}_n) \cdot \nabla.$$

For a smooth scalar observable A_n ,

$$\left. \frac{d^r}{d\tau^r} A_n(\theta_n(\tau)) \right|_{\tau=0} = \mathcal{V}_n^r A_n(\theta_n(0)). \quad (33.1)$$

At fixed r , repeated application of $\mathcal{V}_n$ creates a finite sum of scalar contractions of derivatives of A_n , the network, and the loss at initialization. These are peeling inputs.

34. Hidden feature kernels

For example,

$$K_{n,ab}^\ell(\tau) = \frac{1}{n_\ell} \sum_i h_i^\ell(x^a; \theta_\tau) h_i^\ell(x^b; \theta_\tau).$$

The intended feature-learning calculation is

$$\frac{d^r}{d\tau^r} K_{n,ab}^\ell(0) \xrightarrow{n \rightarrow \infty} \text{a finite sum of products of layerwise Gaussian moments.}$$

The logical chain is

training-time derivative $\longrightarrow$ initialization derivative contractions $\longrightarrow$ peeling states $\longrightarrow$ explicit Gaussian moments.

35. What this does and does not yet establish

The program computes fixed-order initialization jets. It does not by itself prove:

- interchange of the infinite-width limit with time differentiation;
- convergence or concentration of every coefficient;
- a nonzero radius of convergence of the time-Taylor series;
- recovery of a finite-time training trajectory from its local jet;
- validity of a fixed-step learning-rate expansion when the number of steps scales like $1/\eta$.

These are separate analytic obligations.

Part IV. Worked execution: the backward sensitivity kernel

36. Definition and normalization

“Backward kernel” is taken here to mean the output-sensitivity kernel. Let hidden layers be $1, \dots, H$, where $H = L - 1$, all of width n . Use hidden effective weights

$$W_{ij}^\ell \sim N\left(0, \frac{s_\ell^2}{n}\right).$$

Write the scalar readout as

$$f_a = s_L n^{-\alpha} \sum_{i=1}^n a_i h_i^H(a), \quad a_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

The choices

$$\alpha = \frac{1}{2}$$

and

$$\alpha = 1$$

represent, respectively, a standard random-output/NTK-size readout and a common mean-field/ μ P-size readout.

Define

$$g_i^{\ell,a} := \frac{\partial f_a}{\partial z_i^\ell(a)}, \quad \beta_i^{\ell,a} := n^\alpha g_i^{\ell,a}.$$

Then

$$\beta_i^{H,a} = s_L a_i \phi'(z_i^{H,a}), \tag{36.1}$$

and, for $\ell < H$,

$$\beta_i^{\ell,a} = \phi'(z_i^{\ell,a}) \sum_{r=1}^n W_{ri}^{\ell+1} \beta_r^{\ell+1,a}. \tag{36.2}$$

The normalized backward kernel is

$$\widehat{\Pi}_{n,ab}^\ell := \frac{1}{n} \sum_i \beta_i^{\ell,a} \beta_i^{\ell,b} = n^{2\alpha-1} \sum_i g_i^{\ell,a} g_i^{\ell,b}. \tag{36.3}$$

For $\alpha = 1/2$, this is the ordinary backward inner product. For $\alpha = 1$, the ordinary inner product is $n^{-1} \widehat{\Pi}^\ell$.

37. Gaussian derivative coefficients

Let

$$Z^\ell \sim N(0, Q^\ell)$$

and define

$$D_{ab}^\ell := \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell)]. \tag{37.1}$$

For hidden transitions, define

$$\dot{Q}_{ab}^\ell := s_{\ell+1}^2 D_{ab}^\ell.$$

38. Boundary peel

At the last hidden layer,

$$\widehat{\Pi}_{n,ab}^H = \frac{s_L^2}{n} \sum_i a_i^2 \phi'(z_i^{H,a}) \phi'(z_i^{H,b}).$$

The readout variables are independent of z^H , so a weighted law of large numbers gives

$$\widehat{\Pi}_{n,ab}^H \longrightarrow s_L^2 D_{ab}^H. \quad (38.1)$$

This is a pure readout Wick contraction followed by a rowwise Gaussian expectation.

39. Recursive scalarization

Let

$$u_i^{\ell,ab} := \phi'(z_i^{\ell,a}) \phi'(z_i^{\ell,b}).$$

Substituting the backward recursion into the kernel gives the exact scalar expression

$$\widehat{\Pi}_{n,ab}^\ell = \frac{1}{n} \sum_{i,r,s} u_i^{\ell,ab} W_{ri}^{\ell+1} W_{si}^{\ell+1} \beta_r^{\ell+1,a} \beta_s^{\ell+1,b}. \quad (39.1)$$

This is recursive scalarization: only the current weight layer has been exposed, while the upper backward state remains packaged as a function of upper preactivations.

40. Exact finite-width Wick–Stein peel

Set

$$v_{\ell+1} = \frac{s_{\ell+1}^2}{n}.$$

For a smooth function F of $W^{\ell+1}$, Gaussian integration by parts gives

$$\mathbb{E}[W_{ri}^{\ell+1} W_{si}^{\ell+1} F] = v_{\ell+1} \delta_{rs} \mathbb{E}[F] + v_{\ell+1}^2 \mathbb{E}[\partial_{W_{ri}^{\ell+1}} \partial_{W_{si}^{\ell+1}} F]. \quad (40.1)$$

Apply this with

$$F = \beta_r^{\ell+1,a} \beta_s^{\ell+1,b}.$$

Because $W_{ri}^{\ell+1}$ influences the upper network through the batch preactivation vector of neuron r ,

$$\partial_{W_{ri}^{\ell+1}} = \sum_{c=1}^B h_i^{\ell,c} \partial_{z_r^{\ell+1,c}}.$$

Therefore

$$\mathbb{E}_{W^{\ell+1}}[\widehat{\Pi}_{n,ab}^\ell] = s_{\ell+1}^2 \widehat{D}_{n,ab}^\ell \mathbb{E}_{W^{\ell+1}}[\widehat{\Pi}_{n,ab}^{\ell+1}] + R_{n,ab}^\ell, \quad (40.2)$$

where

$$\widehat{D}_{n,ab}^\ell = \frac{1}{n} \sum_i u_i^{\ell,ab}$$

and

$$\begin{aligned} R_{n,ab}^\ell &= \frac{s_{\ell+1}^4}{n^3} \sum_{i,r,s} \sum_{c,d=1}^B u_i^{\ell,ab} h_i^{\ell,c} h_i^{\ell,d} \\ &\quad \times \mathbb{E} \left[\partial_{z_r^{\ell+1,c}} \partial_{z_s^{\ell+1,d}} \left(\beta_r^{\ell+1,a} \beta_s^{\ell+1,b} \right) \right]. \end{aligned} \quad (40.3)$$

Equation (40.2) is exact at finite width after the stated conditional expectation. It displays both the pseudo-independent Wick branch and the Onsager remainder.

41. First nontrivial explicit correction

Consider the transition from layer $H-1$ to layer H . Integrating the centered readout gives

$$\mathbb{E}[a_r a_s] = \delta_{rs}.$$

Let

$$\psi_{ab}(Z) = \phi'(Z_a) \phi'(Z_b).$$

Conditional on $\mathcal{F}_{H-1}$,

$$\text{Cov}(W_{ri}^H, z_r^{H,c} \mid \mathcal{F}_{H-1}) = \frac{s_H^2}{n} h_i^{H-1,c}.$$

Hence

$$\begin{aligned} \mathbb{E}[(W_{ri}^H)^2 \psi_{ab}(Z_r^H) \mid \mathcal{F}_{H-1}] &= \frac{s_H^2}{n} \mathbb{E}[\psi_{ab}(Z)] \\ &+ \frac{s_H^4}{n^2} \sum_{c,d=1}^B h_i^{H-1,c} h_i^{H-1,d} \mathbb{E}[\partial_{cd} \psi_{ab}(Z)], \end{aligned} \quad (41.1)$$

where $Z \sim N(0, Q_n^H)$.

Substitution yields

$$\begin{aligned} \mathbb{E}[\widehat{\Pi}_{n,ab}^{H-1} \mid \mathcal{F}_{H-1}] &= s_H^2 \widehat{D}_{n,ab}^{H-1} \left[s_L^2 \mathbb{E}_{Z \sim N(0, Q_n^H)} [\phi'(Z_a) \phi'(Z_b)] \right] \\ &+ \frac{s_L^4 s_H^4}{n^2} \sum_i u_i^{H-1,ab} \sum_{c,d=1}^B h_i^{H-1,c} h_i^{H-1,d} \mathbb{E}[\partial_{cd} \psi_{ab}(Z)]. \end{aligned} \quad (41.2)$$

The second line is the explicit Onsager correction. It is $O(n^{-1})$ for fixed batch size and bounded moments. For $a \neq b$, its Gaussian integrands are combinations of

$$\phi'''(Z_a) \phi'(Z_b), \quad \phi''(Z_a) \phi''(Z_b), \quad \phi'(Z_a) \phi'''(Z_b).$$

The leading term gives

$$\Pi_{ab}^{H-1} = s_H^2 D_{ab}^{H-1} s_L^2 D_{ab}^H.$$

42. Two-path diagram and leading ladder pattern

Unrolling backpropagation gives

$$\beta_{i_\ell}^{\ell,a} = s_L \sum_{i_{\ell+1}, \dots, i_H} a_{i_H} \left(\prod_{t=\ell}^H \phi'(z_{i_t}^{t,a}) \right) \left(\prod_{t=\ell+1}^H W_{i_t i_{t-1}}^t \right). \quad (42.1)$$

The kernel contains two such paths, one for sample a and one for sample b .

The leading equality pattern is the ladder:

1. The readout contraction forces the two terminal indices to agree.
2. At each lower hidden layer, the weight–weight contraction forces the two path indices to agree.
3. Each shared hidden index contributes n choices.
4. Each paired hidden weight contributes a covariance factor n^{-1} .
5. These powers cancel layer by layer.
6. The paired activation derivatives average to D_{ab}^t .

A Stein attachment may release one of these Wick equalities and produce an extra free index, but it also carries an extra inverse-width covariance. Because the two paths share their bottom index in the kernel definition, any split must eventually reconnect or satisfy an additional equality, losing a width factor.

For a centered readout, a nominally leading disconnected split also vanishes because each isolated backward branch has zero first moment. Thus

$$R_{n,ab}^\ell = O(n^{-1})$$

in expectation for fixed depth under the stated regularity conditions.

This is the precise pseudo-independence result for this observable. It is not a universal statement about all backward or higher-derivative observables.

43. Backward recursion

The deterministic limiting recursion is

$$\Pi^H = s_L^2 D^H$$

and

$$\Pi^\ell = s_{\ell+1}^2 D^\ell \odot \Pi^{\ell+1}, \quad 1 \leq \ell < H, \quad (43.1)$$

where $\odot$ is entrywise multiplication.

Equivalently,

$$\Pi_{ab}^\ell = \left(\prod_{r=\ell+1}^L s_r^2 \right) \left(\prod_{t=\ell}^H D_{ab}^t \right). \quad (43.2)$$

This is a one-state, $O(L)$ execution of the peeling program.

44. Concentration

To pass from the expected recursion to a deterministic kernel limit, expand the second moment. Four backward paths appear. The leading disconnected pair of ladders reproduces

$$(\mathbb{E}\widehat{\Pi}_{n,ab}^\ell)^2.$$

Every connected four-path pattern loses at least one free neuron index. Under standard fixed-depth moment assumptions,

$$\text{Var}(\widehat{\Pi}_{n,ab}^\ell) = O(n^{-1}),$$

so

$$\widehat{\Pi}_{n,ab}^\ell \rightarrow \Pi_{ab}^\ell \quad \text{in } L^2.$$

The dropped deterministic Stein bias is typically $O(n^{-1})$, whereas finite-width random fluctuations are typically $O_p(n^{-1/2})$.

45. Square-loss adjoint

For

$$\mathcal{L}_a = (f_a - y_a)^2, \quad \rho_a = 2(f_a - y_a),$$

the normalized loss adjoint is

$$n^\alpha \frac{\partial \mathcal{L}_a}{\partial z_i^\ell(a)} = \rho_a \beta_i^{\ell,a}.$$

Therefore its backward kernel obeys the exact identity

$$\widehat{\Pi}_{n,ab}^{\ell,\text{loss}} = 4(f_a - y_a)(f_b - y_b)\widehat{\Pi}_{n,ab}^\ell. \quad (45.1)$$

For a centered mean-field readout with $\alpha = 1$, $f_a \rightarrow 0$, giving

$$\widehat{\Pi}_{n,ab}^{\ell,\text{loss}} \rightarrow 4y_a y_b \Pi_{ab}^\ell.$$

For $\alpha = 1/2$, the output has a nondegenerate Gaussian-process limit F , and the loss kernel has the generally random limit

$$4(F_a - y_a)(F_b - y_b)\Pi_{ab}^\ell.$$

If F is centered with covariance K^f , its annealed expectation is

$$4(K_{ab}^f + y_a y_b)\Pi_{ab}^\ell.$$

46. Relation to the NTK sample

The standard backward kernel Π^ℓ is a second-moment object. It should not be conflated with the mixed first moment $M_{k,\ell}$ in the original sample.

With a fully specified parameter coordinate and optimizer metric, the layer- ℓ NTK contribution has the schematic form

$$\text{forward activation Gram at layer } \ell - 1 \quad \odot \quad \Pi^\ell,$$

times the relevant parameter-scaling factor. This distinction explains why the direct NTK term contains a forward activation covariance, while backward transport contributes the derivative factors.

46A. Complete concrete execution through groups 4, 3, 2, and 1

This section records the fully scalarized four-layer example used in the discussion. It uses hidden groups 1, 2, 3 and a scalar readout group 4, all hidden widths being n . There is no extra depth symbol in this calculation.

For two fixed inputs x, x' , let

$$z_p^2(u) = \frac{1}{\sqrt{n}} \sum_a W_{pa}^2 h_a^1(u), \quad z_r^3(u) = \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 h_p^2(u),$$

$$h_i^\ell(u) = \phi(z_i^\ell(u)), \quad z^4(u) = \frac{1}{n} \sum_r W_r^4 h_r^3(u),$$

where every displayed raw weight is independent $N(0, 1)$. Assume $\phi \in C^3$ and that all Gaussian and empirical moments used below are uniformly integrable. Define

$$\delta_i^\ell(u) = n \frac{\partial z^4(u)}{\partial z_i^\ell(u)}, \quad B_n^1(x, x') = \frac{1}{n} \sum_a \delta_a^1(x) \delta_a^1(x').$$

The chain rule gives

$$\delta_r^3(u) = W_r^4 \phi'(z_r^3(u)),$$

$$\delta_p^2(u) = \frac{1}{\sqrt{n}} \phi'(z_p^2(u)) \sum_r W_{rp}^3 \delta_r^3(u),$$

$$\delta_a^1(u) = \frac{1}{\sqrt{n}} \phi'(z_a^1(u)) \sum_p W_{pa}^2 \delta_p^2(u).$$

The exact scalarization is

$$\begin{aligned} B_n^1(x, x') &= \frac{1}{n^3} \sum_{a,p,q,r,t} W_r^4 W_t^4 W_{rp}^3 W_{tq}^3 W_{pa}^2 W_{qa}^2 \\ &\times \phi'(z_a^1(x)) \phi'(z_a^1(x')) \phi'(z_p^2(x)) \phi'(z_p^2(x')) \phi'(z_r^3(x)) \phi'(z_r^3(x')). \end{aligned} \quad (46A.1)$$

All indices in this section are explicit. The index r belongs only to group 3 after the readout is integrated. The indices p, q also occur in group 2 and therefore cannot be summed out during the group-3 peel. The index a occurs in groups 2 and 1 and must remain until the group-1 peel.

The exact rowwise Gaussian identities

Let $w_1, \dots, w_n$ be independent $N(0, 1)$ and

$$Z(u) = \frac{1}{\sqrt{n}} \sum_j w_j h_j(u).$$

Conditionally on the lower-layer numbers $h_j(u)$, Gaussian integration by parts gives

$$\mathbb{E}[w_p G(Z)] = \frac{1}{\sqrt{n}} \sum_u h_p(u) \mathbb{E}[\partial_u G(Z)] \quad (46A.2)$$

and, after applying it twice,

$$\mathbb{E}[w_p w_q G(Z)] = \delta_{pq} \mathbb{E}[G(Z)] + \frac{1}{n} \sum_{u,v} h_p(u) h_q(v) \mathbb{E}[\partial_u \partial_v G(Z)]. \quad (46A.3)$$

Here and below every sample index u, v, s, t is summed over $\{x, x'\}$, and ∂_u means differentiation with respect to the Gaussian coordinate $Z(u)$. Equation (46A.3), not an independence claim, is the exact Wick–Stein rule used at each hidden layer.

Peel group 4

Since the readout weights are independent of all hidden groups,

$$\mathbb{E}[W_r^4 W_t^4] = \delta_{rt}.$$

Thus

$$\begin{aligned} \mathbb{E}_{W^4}[B_n^1] &= \frac{1}{n^3} \sum_{a,p,q,r} W_{rp}^3 W_{rq}^3 W_{pa}^2 W_{qa}^2 \\ &\times \phi'(z_a^1(x)) \phi'(z_a^1(x')) \phi'(z_p^2(x)) \phi'(z_p^2(x')) \phi'(z_r^3(x)) \phi'(z_r^3(x')). \end{aligned} \quad (46A.4)$$

Only r is now a top-group-only index.

Peel group 3

Condition on groups 1 and 2 and apply (46A.3) to row r of W^3 , with

$$G(Z) = \phi'(Z(x)) \phi'(Z(x')).$$

This gives the exact conditional identity

$$\begin{aligned} \mathbb{E}_{W^3}[W_{rp}^3 W_{rq}^3 \phi'(z_r^3(x)) \phi'(z_r^3(x'))] \\ &= \delta_{pq} \mathbb{E}_{Z \sim N(0, Q_n^3)}[\phi'(Z(x)) \phi'(Z(x'))] \\ &+ \frac{1}{n} \sum_{u,v} h_p^2(u) h_q^2(v) \mathbb{E}_{Z \sim N(0, Q_n^3)}[\partial_u \partial_v \{\phi'(Z(x)) \phi'(Z(x'))\}], \end{aligned} \quad (46A.5)$$

where, at finite width, the covariance inside $\mathbb{E}_3$ is

$$Q_n^3(u, v) = \frac{1}{n} \sum_j h_j^2(u) h_j^2(v).$$

The four possible second derivatives in (46A.5) are

(u, v)	$\partial_u \partial_v \{\phi'(Z(x)) \phi'(Z(x'))\}$
(x, x)	$\phi'''(Z(x)) \phi'(Z(x'))$
(x, x')	$\phi''(Z(x)) \phi''(Z(x'))$
(x', x)	$\phi''(Z(x)) \phi''(Z(x'))$
(x', x')	$\phi'(Z(x)) \phi'''(Z(x'))$

(46A.6)

The factors $h_p^2(u), h_q^2(v)$ created by Stein differentiation belong to group 2, not group 3. They must therefore be retained for the next peel.

For the leading mean-field theorem, replace Q_n^3 by its deterministic forward limit

$$Q^3(u, v) = \mathbb{E}_2[\phi(Z(u))\phi(Z(v))].$$

This is justified by the fixed-batch forward law of large numbers, continuity of the displayed Gaussian expectations in the covariance, and uniform integrability. Because the singled-out rows p, q also occur in Q_n^3 , a fully quantitative proof uses a leave-two-out Gram matrix. The removal changes the covariance by $O(n^{-1})$, while the leave-two-out matrix converges to Q^3 and is conditionally independent of rows p, q .

For compactness only within this execution, write

$$D_\ell = \mathbb{E}_\ell[\phi'(Z(x))\phi'(Z(x'))]$$

and

$$C_\ell(u, v) = \mathbb{E}_\ell[\partial_u \partial_v \{\phi'(Z(x))\phi'(Z(x'))\}],$$

where $\mathbb{E}_\ell$ uses the deterministic covariance Q^ℓ . Substitution of (46A.5) into (46A.4) gives two group-2 expressions:

$$\frac{D_3}{n^2} \sum_{a,p} (W_{pa}^2)^2 \phi'(z_a^1(x)) \phi'(z_a^1(x')) \phi'(z_p^2(x)) \phi'(z_p^2(x')) \quad (46A.7)$$

and

$$\begin{aligned} & \frac{1}{n^3} \sum_{a,p,q} W_{pa}^2 W_{qa}^2 \phi'(z_a^1(x)) \phi'(z_a^1(x')) \\ & \times \sum_{u,v} C_3(u, v) \phi'(z_p^2(x)) \phi(z_p^2(u)) \phi'(z_q^2(x')) \phi(z_q^2(v)). \end{aligned} \quad (46A.8)$$

Peel group 2: the branch from the group-3 Wick pairing

Condition on group 1. The exact conditional covariance of a row of z^2 is

$$Q_n^2(s, t) = \frac{1}{n} \sum_a h_a^1(s) h_a^1(t).$$

For the leading mean-field theorem, replace it by

$$Q^2(s, t) = \mathbb{E}_1[\phi(Z(s))\phi(Z(t))].$$

The group-1 rowwise law of large numbers gives $Q_n^2 \rightarrow Q^2$. Continuity of the Gaussian expectations and uniform integrability allow the replacement inside expectations. A leave-one-out covariance separates a singled-out row a when a quantitative proof requires it.

The exact Gaussian integration-by-parts formula has the form below with covariance Q_n^2 . After the stated deterministic-covariance replacement, applying (46A.3) to (46A.7) gives

$$\begin{aligned} & \mathbb{E}_{W^2}[(W_{pa}^2)^2 \phi'(z_p^2(x)) \phi'(z_p^2(x'))] \\ & = D_2 + \frac{1}{n} \sum_{s,t} h_a^1(s) h_a^1(t) C_2(s, t). \end{aligned} \quad (46A.9)$$

After summing the top-group-only index p , (46A.7) becomes

$$\frac{D_3 D_2}{n} \sum_a \phi'(z_a^1(x)) \phi'(z_a^1(x')) \quad (46A.10)$$

plus

$$\frac{D_3}{n^2} \sum_a \phi'(z_a^1(x)) \phi'(z_a^1(x')) \sum_{s,t} h_a^1(s) h_a^1(t) C_2(s, t). \quad (46A.11)$$

Peel group 2: the branch created by group-3 Stein differentiation

The equality pattern of p, q must be split before rowwise independence is used.

For $p = q$, define only for the next two displays

$$P_{uv}(Z) = \phi'(Z(x)) \phi'(Z(x')) \phi(Z(u)) \phi(Z(v)).$$

Then (46A.3) gives

$$\mathbb{E}_{W^2}[(W_{pa}^2)^2 P_{uv}(z_p^2)] = \mathbb{E}_2[P_{uv}(Z)] + \frac{1}{n} \sum_{s,t} h_a^1(s) h_a^1(t) \mathbb{E}_2[\partial_s \partial_t P_{uv}(Z)]. \quad (46A.12)$$

After summing p , the diagonal branch of (46A.8) becomes

$$\frac{1}{n^2} \sum_a \phi'(z_a^1(x)) \phi'(z_a^1(x')) \sum_{u,v} C_3(u, v) \mathbb{E}_2[P_{uv}(Z)] \quad (46A.13)$$

plus

$$\frac{1}{n^3} \sum_a \phi'(z_a^1(x)) \phi'(z_a^1(x')) \sum_{u,v,s,t} C_3(u,v) h_a^1(s) h_a^1(t) \times \mathbb{E}_2[\partial_s \partial_t P_{uv}(Z)]. \quad (46A.14)$$

For $p \neq q$, the two rows of W^2 are conditionally independent. Set

$$L_u(Z) = \phi'(Z(x)) \phi(Z(u)), \quad R_v(Z) = \phi'(Z(x')) \phi(Z(v)).$$

Using the one-weight identity (46A.2) on each row,

$$\mathbb{E}_{W^2}[W_{pa}^2 L_u(z_p^2)] = \frac{1}{\sqrt{n}} \sum_s h_a^1(s) \mathbb{E}_2[\partial_s L_u(Z)], \quad (46A.15)$$

$$\mathbb{E}_{W^2}[W_{qa}^2 R_v(z_q^2)] = \frac{1}{\sqrt{n}} \sum_t h_a^1(t) \mathbb{E}_2[\partial_t R_v(Z)]. \quad (46A.16)$$

There are $n(n-1)$ ordered pairs $p \neq q$. Their total contribution is

$$\frac{n-1}{n^3} \sum_a \phi'(z_a^1(x)) \phi'(z_a^1(x')) \sum_{u,v,s,t} C_3(u,v) h_a^1(s) h_a^1(t) \times \mathbb{E}_2[\partial_s L_u(Z)] \mathbb{E}_2[\partial_t R_v(Z)]. \quad (46A.17)$$

The derivatives in these equations are completely explicit. For example,

$$\partial_s L_u(Z) = \mathbf{1}_{s=x} \phi''(Z(x)) \phi(Z(u)) + \mathbf{1}_{s=u} \phi'(Z(x)) \phi'(Z(u)), \quad (46A.18)$$

and the analogous formula for R_v replaces x by x' . To expand the remaining second derivative without hiding any product-rule terms, regard P_{uv} as four factor occurrences with labels and derivative orders

$$(x, 1), \quad (x', 1), \quad (u, 0), \quad (v, 0).$$

If these four pairs are denoted locally by (a_j, m_j) , with $\phi^{(0)} = \phi$, then

$$\begin{aligned} \partial_s \partial_t P_{uv}(Z) &= \sum_{j=1}^4 \mathbf{1}_{a_j=s} \mathbf{1}_{a_j=t} \phi^{(m_j+2)}(Z(a_j)) \prod_{k \neq j} \phi^{(m_k)}(Z(a_k)) \\ &\quad + \sum_{j \neq k} \mathbf{1}_{a_j=s} \mathbf{1}_{a_k=t} \phi^{(m_j+1)}(Z(a_j)) \phi^{(m_k+1)}(Z(a_k)) \prod_{l \neq j,k} \phi^{(m_l)}(Z(a_l)). \end{aligned} \quad (46A.19)$$

The first sum differentiates one factor occurrence twice; the second differentiates two distinct occurrences. Repeated sample labels, such as $u = x$, are handled correctly because the occurrences remain distinct in the product rule.

Peel group 1

After the group-2 peel, every remaining random factor is a function of one group-1 row. There is no explicit W^1 factor in this observable because the kernel differentiates with respect to z^1 , not with respect to W^1 . Consequently group 1 is removed by the rowwise law of large numbers, not by another Stein differentiation.

Let $Z \sim N(0, Q^1)$, where Q^1 is the deterministic covariance induced by the two fixed inputs. Then, for every integrable function G occurring above,

$$\frac{1}{n} \sum_a G(z_a^1(x), z_a^1(x')) \longrightarrow \mathbb{E}_1[G(Z(x), Z(x'))]. \quad (46A.20)$$

After the deterministic replacements of Q_n^3 and Q_n^2 , applying (46A.20) to (46A.10)–(46A.17) gives the complete five-branch expression

$$\begin{aligned} \mathbb{E}[B_n^1] &= D_1 D_2 D_3 \\ &\quad + \frac{D_3}{n} \sum_{s,t} C_2(s,t) \mathbb{E}_1[\phi'(Z(x)) \phi'(Z(x')) \phi(Z(s)) \phi(Z(t))] \\ &\quad + \frac{D_1}{n} \sum_{u,v} C_3(u,v) \mathbb{E}_2[P_{uv}(Z)] \\ &\quad + \frac{1}{n^2} \sum_{u,v,s,t} C_3(u,v) \mathbb{E}_2[\partial_s \partial_t P_{uv}(Z)] \\ &\quad \quad \times \mathbb{E}_1[\phi'(Z(x)) \phi'(Z(x')) \phi(Z(s)) \phi(Z(t))] \\ &\quad + \frac{n-1}{n^2} \sum_{u,v,s,t} C_3(u,v) \mathbb{E}_2[\partial_s L_u(Z)] \mathbb{E}_2[\partial_t R_v(Z)] \\ &\quad \quad \times \mathbb{E}_1[\phi'(Z(x)) \phi'(Z(x')) \phi(Z(s)) \phi(Z(t))] + o(1). \end{aligned} \quad (46A.21)$$

The $o(1)$ records the two deterministic-covariance replacements. Equation (46A.21) is therefore a complete structural expansion after that mean-field reduction, not an exact finite-width $1/n$ asymptotic expansion. In particular, fluctuations of Q_n^2, Q_n^3 can alter the displayed $1/n$ coefficient even

though they cannot alter the leading limit.

The five displayed branches have orders

$$1, \quad n^{-1}, \quad n^{-1}, \quad n^{-2}, \quad n^{-1},$$

respectively. Thus every branch containing at least one Stein attachment vanishes in the leading fixed-depth mean-field expectation, while none was discarded before its index multiplicity was counted. The final result is

$$\lim_{n \rightarrow \infty} \mathbb{E}[B_n^1(x, x')] = D_1 D_2 D_3. \quad (46A.22)$$

This proves the expectation limit under the stated smoothness, moment, deterministic-forward-Gram, and uniform-integrability assumptions. Upgrading (46A.22) to convergence of B_n^1 itself requires the separate four-path second-moment peel described in Section 44.

Lessons for the peeling lemma

This execution makes the following refinements necessary.

1. A peel is an exact conditional Gaussian elimination followed by a separate mean-field replacement; these are not the same step.
2. Only indices whose remaining occurrences all belong to the current top group may be summed out during that peel.
3. Lower-group factors created by Stein differentiation must remain as boundary data for the next peel.
4. Equality patterns such as $p = q$ and $p \neq q$ must be separated before independence and width counting are used.
5. Deterministic Gram replacement in an expression containing singled-out rows requires joint convergence or a leave-finitely-many-out argument, plus uniform integrability for expectations.
6. A leading-order theorem may replace empirical Grams by their limits. An exact $1/n$ expansion must retain Gram fluctuations and their correlations.
7. The final group need not contain an explicit Gaussian weight. In this example it is peeled by a rowwise Gaussian law of large numbers.
8. Expectation convergence and concentration of the observable are distinct proof obligations.

Part V. Small reusable examples

47. Equality-pattern example

Let Z_i be iid copies of Z . Then

$$\frac{1}{n^2} \sum_{i,j} F(Z_i)G(Z_j) = \frac{n(n-1)}{n^2} \left(\frac{1}{n(n-1)} \sum_{i \neq j} F(Z_i)G(Z_j) \right) + \frac{1}{n^2} \sum_i F(Z_i)G(Z_i).$$

The off-diagonal partition has two free row blocks and converges to

$$\mathbb{E}[F(Z)]\mathbb{E}[G(Z)].$$

The diagonal partition has one free row block and is lower by n^{-1} under this normalization. If the observable instead contains $n^{-1} \sum_i F(Z_i)G(Z_i)$, the diagonal partition is leading and converges to the joint moment.

This is the basic equality-partition mechanism behind a peel.

48. One-weight Stein example

Let (X, Z) be centered jointly Gaussian. Then

$$\mathbb{E}[XF(Z)] = \text{Cov}(X, Z)\mathbb{E}[F'(Z)].$$

The explicit weight disappears, one derivative is added to the nonlinear function, and the covariance inserts its lower-layer factor.

49. Two-weight Wick–Stein example

For centered jointly Gaussian (X, Y, Z) ,

$$\mathbb{E}[XYF(Z)] = \text{Cov}(X, Y)\mathbb{E}[F(Z)] + \text{Cov}(X, Z)\text{Cov}(Y, Z)\mathbb{E}[F''(Z)].$$

The first term is the pseudo-independent Wick branch. The second is the Onsager branch. In a network, their relative size is determined only after the resulting index constraints and lower-layer activations are summed.

Part VI. Open proof and development program

50. Immediate formalization tasks

1. Choose one canonical μP parameterization and optimizer metric.
2. Define the admissible contraction-graph grammar.
3. Define state equivalence and canonicalization.
4. Prove the generalized conditional Wick–Stein elimination lemma with all row and batch indices.
5. Define a global width-degree valuation for diagrams.
6. Prove finite-state closure for selected observable families.
7. Establish uniform moment bounds under the required activation smoothness.
8. Prove covariance replacement $Q_n^\ell \rightarrow Q^\ell$ for retained states.
9. Separate expectation, concentration, and fluctuation theorems.
10. Extend the formalism to nonsmooth activations.

51. Worked-example status and next executions

The companion `MUPTRAININGPEELINGCASESTUDY.md`⁹ now executes:

1. the forward covariance recursion used by the training calculation;
2. the backward sensitivity kernels and resulting μ P tangent kernel;
3. the one-step loss coefficient and hidden-Gram linear cancellation;
4. the one-step quadratic hidden-Gram coefficient, including a surviving off-diagonal fluctuation branch;
5. the exact two-step Euler/Hessian decomposition and readout peel;
6. a formal multichannel nonlinear recursion schema for the two-step correction;
7. a complete deep-linear audit of the one-step and two-step coefficients.

The next required executions and proofs are:

1. a complete nonlinear channel registry for the two-step correction;
2. a joint conditional-CLT theorem for the registered Gaussian base channels;
3. a complete four-path concentration peel for the backward kernels;
4. a standard NTK recursion with every backward-dependence correction exposed;
5. a fluctuation-scale observable requiring covariance expansion and connected diagrams;
6. a nonsmooth-activation execution using weak derivatives or Gaussian boundary terms.

52. Questions to keep explicit

- Which quantities are deterministic in the limit and which retain Gaussian randomness?
- Which readout centering or zero-initialization convention is used?
- Are derivatives taken with respect to raw or effective weights?
- Which Stein branches survive under the selected normalization?
- Does the recursive state space actually close independently of depth?
- What activation derivative order is generated by the complete peel?
- Is the claim about an annealed expectation, convergence in probability, or a fluctuation law?
- Are finite-time feature dynamics being claimed, or only their initialization jet?

Change log

2026-08-03 — Consolidated μ P training case study

- Added the companion `MUPTRAININGPEELINGCASESTUDY.md`¹⁰.
- Consolidated the backward-kernel peel, one-step linear loss and quadratic hidden-Gram calculations, and the two-step gradient-correction calculation in one raw-coordinate μ P convention.
- Recorded the new program-level lessons: exact conditional Stein integration before deterministic Gram replacement, separate concentration obligations, survival of off-diagonal families after introducing a second copy, and multichannel derivative-state augmentation at later training steps.
- Updated the worked-example roadmap to distinguish completed calculations, formal nonlinear schemas, and remaining theorem obligations.

2026-08-01 — Initial living Markdown document

- Preserved the original mean-field peeling proposal in Part I.
- Replaced global Gaussianity with exact layerwise conditional Gaussianity in the maintained interpretation.
- Replaced informal independent-copy language with the generalized Wick–Stein identity.
- Formalized scalar contraction states, equality partitions, global width power counting, and dynamic-programming closure.
- Added the gradient-flow feature-learning interface.
- Added a complete first execution on the normalized backward sensitivity kernel, including its explicit first Onsager correction, leading ladder pattern, limiting recursion, concentration statement, and square-loss extension.
- Recorded theorem obligations and next worked examples.
- Standardized Markdown mathematics to inline and display dollar delimiters for renderer compatibility.

Appendix A. Original NTK sample execution

This appendix preserves the mathematical content of the supplied LaTeX sample in Markdown. The document preamble and package declarations are omitted, but its claims, definitions, recursions, proof sketches, and conclusions are retained as a source record. The notation in this appendix is not silently corrected; Parts II–IV contain the maintained versions.

A.1 Title and stated purpose

Neural Tangent Kernel: A First Principles Derivation

Amir Joudaki, ETH Zurich

The sample described itself as a first-principles derivation of the Neural Tangent Kernel using the law of large numbers, central limit theorem, and Gaussian integration identities. It aimed to:

- establish mean-field variance recursions;
- analyze kernel shifts through parameter updates;
- derive the classical NTK recursion;
- analyze possible off-diagonal backward interactions;
- argue that these corrections are negligible at infinite width.

A.2 Network architecture in the sample

The sample used an input layer, $L - 1$ hidden layers, and a scalar output:

$$h^0(x) = x \in \mathbb{R}^{d_0},$$

⁹<MUP_TRAINING_PEELING_CASE_STUDY.md>

¹⁰<MUP_TRAINING_PEELING_CASE_STUDY.md>

$$z^\ell(x) = W^{\ell-1}h^{\ell-1}(x) + b^{\ell-1},$$

$$h^\ell(x) = \phi(z^\ell(x)), \quad 1 \leq \ell < L,$$

and

$$z^L(x) = W^{L-1}h^{L-1}(x) + b^{L-1}.$$

Weights and biases were initialized as

$$W_{ij}^\ell \sim N\left(0, \frac{\sigma_w^2}{d_\ell}\right), \quad b_i^\ell \sim N(0, \sigma_b^2).$$

For the sample derivation, the “loss” was changed from the squared loss in the baseline notes to the scalar output

$$\mathcal{L} = z^L(x').$$

The stated motivation was to isolate the gradient structure and obtain a simple first-moment boundary.

A.3 Forward kernel recursion in the sample

The preactivation kernel was defined by

$$\Sigma^\ell(x, x') := \frac{1}{d_\ell} \langle z^\ell(x), z^\ell(x') \rangle.$$

The proposed infinite-width recursion was

$$\Sigma^0(x, x') := \sigma_w^2 \frac{x^\top x'}{d_0} + \sigma_b^2,$$

$$\Sigma^{\ell+1}(x, x') := \sigma_w^2 \mathbb{E}[\phi(z)\phi(z')] + \sigma_b^2,$$

where (z, z') is centered Gaussian with covariance

$$\begin{pmatrix} \Sigma^\ell(x, x) & \Sigma^\ell(x, x') \\ \Sigma^\ell(x, x') & \Sigma^\ell(x', x') \end{pmatrix}.$$

The derivative kernel was defined as

$$\dot{\Sigma}^\ell(x, x') := \sigma_w^2 \mathbb{E}[\phi'(z)\phi'(z')],$$

with the Gaussian covariance inherited from the previous forward layer.

A.4 Influence function in the sample

For a parameter block W^{k-1} , the sample defined the influence on layer ℓ by

$$\dot{\Theta}_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \left\langle \frac{\partial z_i^\ell(x)}{\partial W^{k-1}}, \frac{\partial z^L(x')}{\partial W^{k-1}} \right\rangle_{W^{k-1}}.$$

Using the derivative of $W a$, it rewrote this schematically as

$$\dot{\Theta}_{k,\ell} = \langle h^{k-1}(x), h^{k-1}(x') \rangle \frac{1}{d_\ell} \sum_i \left\langle \frac{\partial z_i^\ell(x)}{\partial z^k}, \frac{\partial z^L(x')}{\partial z^k} \right\rangle_{z^k}.$$

A.5 Backward Jacobian moments in the sample

With

$$J_j^{(\ell \leftarrow k)}(x) := \frac{\partial z^\ell(x)}{\partial z_j^k},$$

the sample introduced

$$M_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \mathbb{E} \left[\sum_{j=1}^{d_k} J_{j,i}^{(\ell \leftarrow k)}(x) J_j^{(L \leftarrow k)}(x') \right],$$

$$Q_{k,\ell} := \frac{1}{d_\ell} \sum_{i=1}^{d_\ell} \mathbb{E} \left[\left(\sum_{j=1}^{d_k} J_{j,i}^{(\ell \leftarrow k)}(x) J_j^{(L \leftarrow k)}(x') \right)^2 \right],$$

and

$$V_{k,\ell} = Q_{k,\ell} - M_{k,\ell}^2.$$

A.6 First-moment recursion proposed in the sample

The sample stated

$$M_{k,\ell} = \dot{\Sigma}^k(x, x') M_{k+1,\ell},$$

and therefore

$$M_{k,\ell} = M_{\ell,\ell} \prod_{i=k}^{\ell-1} \dot{\Sigma}^i(x, x').$$

Its proof expanded the Jacobian recursion

$$\frac{\partial z_i^{\ell+1}}{\partial z_j^\ell} = W_{ij}^\ell \phi'(z_j^\ell)$$

inside $M_{k,\ell}$, producing a term of the form

$$M_{k,\ell} = \frac{1}{d_\ell} \sum_{i,j,s,r} \mathbb{E} \left[\frac{\partial z_i^\ell(x)}{\partial z_s^{k+1}} \frac{\partial z_r^L(x')}{\partial z_r^{k+1}} \times \phi'(z_j^k(x)) \phi'(z_j^k(x')) W_{sj}^k W_{rj}^k \right].$$

The source proof then invoked independence of the backward Jacobians from W^k and used

$$\mathbb{E}[W_{sj}^k W_{rj}^k] = \delta_{sr} \frac{\sigma_w^2}{d_k}.$$

It concluded with a derivative-kernel factor times the next backward moment.

The maintained document does not adopt this independence step without a Wick–Stein analysis, because upper activation derivatives can contain preactivations depending on W^k .

A.7 Boundary conditions proposed in the sample

The sample stated

$$M_{\ell,\ell} = \begin{cases} 0, & \ell < L, \\ 1, & \ell = L. \end{cases}$$

The reasoning was that

$$M_{\ell,\ell} = \frac{1}{d_\ell} \sum_i \mathbb{E} \left[\frac{\partial z_i^L(x')}{\partial z_i^\ell} \right],$$

which equals one at the scalar output and was asserted to vanish at hidden layers by independence and symmetry.

The resulting proposed complete first moment was

$$M_{k,\ell} = \begin{cases} 0, & \ell < L, \\ \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x'), & \ell = L. \end{cases}$$

A.8 Second-moment statement in the sample

The sample proposed

$$Q_{k,\ell} = \left(\dot{\Sigma}^k(x, x') \right)^2 Q_{k+1,\ell} + O(1/d_k),$$

with boundary relation

$$Q_{\ell,\ell} = Q_{\ell+1,\ell+1} \dot{\Sigma}^{\ell+1}(x', x') \frac{d_{\ell+1}}{d_\ell}.$$

The proof was described as following from an analogous Wick expansion, with the leading pairing producing the squared derivative-kernel term. The source did not completely define or prove all normalization and boundary details.

A.9 NTK expression in the sample

For a learning-rate scaling of the form $\eta_k \propto \eta_0/d_{k-1}$, the source proposed the expected layer contribution

$$\mathbb{E}[\dot{\Theta}_{k,L}] = \eta_0 \Sigma^{k-1}(x, x') \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x').$$

Summing parameter blocks gave

$$\sum_{k=1}^L \mathbb{E}[\dot{\Theta}_{k,L}] = \eta_0 \sum_{k=1}^L \Sigma^{k-1}(x, x') \prod_{i=k}^{L-1} \dot{\Sigma}^i(x, x').$$

The NTK recursion was then stated as

$$\Theta^L(x, x') = \Theta^{L-1}(x, x') \dot{\Sigma}^{L-1}(x, x') + \Sigma^{L-1}(x, x'),$$

with base case

$$\Theta^1(x, x') = \Sigma^0(x, x').$$

A.10 Signal propagation remark

For $x = x'$, the source wrote

$$\dot{\Sigma}^\ell(x, x) = \sigma_w^2 \mathbb{E}[\phi'(Z)^2],$$

and observed that the product of derivative kernels controls backward signal propagation. It identified the edge-of-chaos condition with

$$\chi_1 := \sigma_w^2 \mathbb{E}[\phi'(\sqrt{q^\star} Z)^2] \approx 1.$$

A.11 Extended Stein identities in the sample

For jointly Gaussian (X, Y, Z) , the source recorded identities of the form

$$\mathbb{E}[f(X)Y] = \sigma_{XY} \mathbb{E}[f'(X)],$$

and

$$\mathbb{E}[f(X)f(Y)Z] = \sigma_{XZ} \mathbb{E}[f'(X)f(Y)] + \sigma_{YZ} \mathbb{E}[f(X)f'(Y)].$$

These were intended to control forward–backward dependence missed by naive independence.

A.12 Tentative diagonal/off-diagonal system

The source proposed a coupled system

$$D_{k,\ell} = \dot{\Sigma}^{k+1} D_{k+1,\ell} + \frac{\sigma_w^4}{d_k} (\mu_{\phi''}^{k+1})^2 C_{\phi,\phi}^k S_{k+1,\ell},$$

$$S_{k,\ell} = 2\sigma_w^4 C_{\phi'',\phi''}^{k+1} (\mu_{\phi}^{k+1})^2 D_{k+1,\ell} + (\mu_{\phi}^{k+1})^2 (\mu_{\phi''}^{k+1})^2 S_{k+1,\ell}.$$

Here D was intended to represent diagonal backward contributions and S off-diagonal corrections. The source claimed:

1. off-diagonal corrections are $O(1/d_k)$ or carry activation-mean factors;
2. centered activations make some off-diagonal terms vanish;
3. the classical NTK is recovered at leading order.

These equations remain archival and tentative because their normalization, state definitions, and boundary conditions were not fully specified.

A.13 Conclusions stated in the sample

The sample concluded that:

1. infinite width produces clean law-of-large-numbers recursions;
2. only the last layer contributes to the chosen first moment under the linear-output surrogate loss;
3. variance contributions vanish at infinite width;
4. off-diagonal backward interactions are negligible;
5. the NTK emerges from parameter-gradient inner products.

It also emphasized that the NTK is independent of the learning rate as an architectural kernel, while the learning rate enters the training update

$$\Delta f = -\eta \Theta \partial_f \mathcal{L}.$$

Finally, it suggested extension to residual networks, normalization layers, and other architectures by modifying the forward and backward recursions.

A.14 Current archival status

This source execution remains important because it supplied the motivating example for the peeling program. It should not be cited as a completed proof without repairing:

- the inconsistent layer indexing;
- the change of loss;
- raw versus effective weight normalization;
- the dependence of upper backward Jacobians on the current weight layer;
- the definitions and normalization of the second-moment states;
- the tentative off-diagonal recursion.

The checked backward-kernel execution in Part IV is the current replacement for the corresponding independence-based step.

Part III. Audited Worked Case Study

Case Study: Mean-Field Peeling for a μP Network at Initialization, After One Step, and After Two Steps

Status: Maintained worked execution

Date: 2026-08-03

Role: A self-contained, low-level execution of the mean-field peeling program on one fixed network and three increasingly difficult observables.

This case study performs three calculations in one notation:

1. the initialization backward-sensitivity kernels;
2. the output and loss changes after one gradient-descent step through order η , and the hidden-Gram change through order η^2 ;
3. the hidden-Gram changes after two gradient-descent steps, through order η^2 , including the correction caused by updating the gradient itself.

The purpose is not merely to quote the answers. Every calculation is organized in the order required by mean-field peeling:

1. fix the raw/effective parameter and optimizer scalings;
2. expand derivatives and training updates before taking a width limit;
3. scalarize the highest active group;
4. peel the readout first;
5. apply exact conditional Wick–Stein identities to the highest hidden group;
6. retain every lower-group factor created by Stein differentiation;
7. split equality patterns before width counting;
8. replace normalized empirical averages by deterministic Gaussian expectations only after all indices using them have been exposed;
9. repeat downward through the network;
10. audit the result by parity, scaling, and a solvable deep-linear specialization.

The finite-width Taylor and chain-rule identities below are exact. The mean-field limits require the regularity assumptions stated in Section 2.7.

Document map

- Section 1¹¹ states the consolidated answers.
- Section 2¹² fixes one raw-coordinate μP convention.
- Section 3¹³ derives the Gaussian identities.
- Section 4¹⁴ executes the backward-kernel peel.
- Section 5¹⁵ computes the one-step coefficients.
- Section 6¹⁶ computes the two-step gradient correction.
- Section 7¹⁷ records amendments to the general program.

Claim-status convention

- **Exact finite-width identity** means no width limit or independence approximation has been used.
- **Mean-field result** means the coefficient follows from the displayed peel under the smoothness, moment, leave-one-out, and concentration assumptions in Section 2.7.
- **Formal mean-field closure** means the finite state and every required Wick–Stein operation have been specified, but a joint convergence theorem for the complete state has not yet been supplied. The general nonlinear two-step recursion has this status; its deep-linear specialization is explicitly audited.

1. Headline results

For the network defined below, let

$$G_{n,ab}^\ell = \frac{1}{n} \sum_i h_i^\ell(a) h_i^\ell(b), \quad \ell = 1, 2, 3,$$

be the hidden activation Gram between batch samples a and b .

At initialization the normalized backward kernels converge to

$$\Pi_{ab}^3 = D_{ab}^3,$$

$$\Pi_{ab}^2 = D_{ab}^2 D_{ab}^3,$$

$$\Pi_{ab}^1 = D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

For one gradient-descent step, the expected sample loss has the linear expansion

$$\mathbb{E}[(f_a^+ - y_a)^2] = y_a^2 - \frac{4\eta}{B} y_a \sum_b K_{ab} y_b + o(\eta) + o_n(1),$$

where

$$K_{ab} = G_{ab}^3 + G_{ab}^2 D_{ab}^3 + G_{ab}^1 D_{ab}^2 D_{ab}^3 + G_{ab}^0 D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

The expected hidden-Gram linear coefficient vanishes at infinite width:

¹¹<#1-headline-results>

¹²<#2-fixed-network-batch-initialization-and- μP -update>

¹³<#3-the-two-exact-gaussian-identities-used-in-every-peel>

¹⁴<#4-execution-i-backward-sensitivity-kernels>

¹⁵<#5-execution-ii-one-gradient-descent-step>

¹⁶<#6-execution-iii-two-gradient-descent-steps>

¹⁷<#7-what-this-case-study-changes-in-the-peeling-program>

$$\lim_{n \rightarrow \infty} \frac{d}{d\eta} \mathbb{E}[G_{n,ab}^{\ell,+}] \Big|_{\eta=0} = 0.$$

Its first generic deterministic coefficient is quadratic:

$$\mathbb{E}[G_{n,ab}^{\ell,+}] = G_{ab}^{\ell} + \eta O(n^{-1}) + \eta^2 C_{ab}^{\ell} + o(\eta^2) + o_n(1).$$

After two Euler gradient-descent steps,

$$\mathbb{E}[G_{n,ab}^{\ell,(2)}] = G_{ab}^{\ell} + \eta O(n^{-1}) + \eta^2 (4C_{ab}^{\ell} + R_{ab}^{\ell}) + o(\eta^2) + o_n(1),$$

where R^{ℓ} is the new gradient-correction term. The change produced by the second update alone is

$$\mathbb{E}[G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)}] = \eta O(n^{-1}) + \eta^2 (3C_{ab}^{\ell} + R_{ab}^{\ell}) + o(\eta^2) + o_n(1).$$

Neither two-step quantity has a deterministic $O(\eta)$ hidden-Gram term.

2. Fixed network, batch, initialization, and μ P update

2.1 Indices

- Batch indices are $a, b, c, d \in \{1, \dots, B\}$.
- First-hidden-layer neuron indices are i, j .
- Second-hidden-layer neuron indices are p, q .
- Third-hidden-layer neuron indices are r, t .
- The input dimension d_0 and batch size B are fixed while $n \rightarrow \infty$.
- Every hidden layer has width n .

2.2 Forward pass

There are three hidden layers and one scalar output. Biases are omitted.

$$\begin{aligned} z_i^1(a) &= \frac{1}{\sqrt{d_0}} \sum_{j=1}^{d_0} W_{ij}^1 x_{aj}, & h_i^1(a) &= \phi(z_i^1(a)), \\ z_p^2(a) &= \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 h_i^1(a), & h_p^2(a) &= \phi(z_p^2(a)), \\ z_r^3(a) &= \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 h_p^2(a), & h_r^3(a) &= \phi(z_r^3(a)), \end{aligned}$$

and

$$f_a = \frac{1}{n} \sum_r v_r h_r^3(a).$$

The raw parameters are independent standard Gaussians:

$$W_{ij}^1, W_{pi}^2, W_{rp}^3, v_r \stackrel{\text{iid}}{\sim} N(0, 1).$$

2.3 Effective parameter variances and learning rates

The effective tensors appearing without explicit normalization would be

$$\frac{W^1}{\sqrt{d_0}}, \quad \frac{W^2}{\sqrt{n}}, \quad \frac{W^3}{\sqrt{n}}, \quad \frac{v}{n}.$$

Their initialization variances and equivalent gradient learning rates are

Effective tensor	Entry variance	Learning rate multiplying its gradient
$W^1/\sqrt{d_0}$	$1/d_0$	$n\eta/d_0$
$W^2/\sqrt{n}$	$1/n$	η
$W^3/\sqrt{n}$	$1/n$	η
v/n	$1/n^2$	η/n

Equivalently, in the raw coordinates used throughout this case study, every block is updated by

$$\theta^+ = \theta - n\eta \nabla_{\theta} \mathcal{J}, \quad \theta \in \{W^1, W^2, W^3, v\}.$$

The fixed d_0 factors are a convention. The displayed width powers are the μ P/maximal-update powers.

2.4 Loss and residuals

Use the average squared batch loss

$$\mathcal{J} = \frac{1}{B} \sum_{c=1}^B (f_c - y_c)^2.$$

Write

$$e_c = f_c - y_c.$$

No factor $1/2$ is placed in the squared loss, so every gradient contains the factor $2/B$.

2.5 Forward activation Grams

Define

$$G_{ab}^0 = \frac{x_a^\top x_b}{d_0},$$

and, at finite width,

$$G_{n,ab}^\ell = \frac{1}{n} \sum_i h_i^\ell(a) h_i^\ell(b), \quad \ell = 1, 2, 3.$$

The deterministic forward recursion is as follows. Let

$$Z^1 \sim N(0, G^0),$$

and define

$$G_{ab}^1 = \mathbb{E}[\phi(Z_a^1) \phi(Z_b^1)].$$

Next let

$$Z^2 \sim N(0, G^1),$$

and define

$$G_{ab}^2 = \mathbb{E}[\phi(Z_a^2) \phi(Z_b^2)].$$

Finally let

$$Z^3 \sim N(0, G^2),$$

and define

$$G_{ab}^3 = \mathbb{E}[\phi(Z_a^3) \phi(Z_b^3)].$$

For each layer define the derivative kernel

$$D_{ab}^\ell = \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell)].$$

Every expectation in these definitions is over one B -dimensional centered Gaussian vector with the displayed covariance.

2.6 Normalized backward variables

Define

$$\delta_i^\ell(a) := n \frac{\partial f_a}{\partial z_i^\ell(a)}.$$

The chain rule gives

$$\delta_r^3(a) = v_r \phi'(z_r^3(a)),$$

$$\delta_p^2(a) = \phi'(z_p^2(a)) \frac{1}{\sqrt{n}} \sum_r W_{rp}^3 \delta_r^3(a),$$

and

$$\delta_i^1(a) = \phi'(z_i^1(a)) \frac{1}{\sqrt{n}} \sum_p W_{pi}^2 \delta_p^2(a).$$

The normalization makes every $\delta_i^\ell(a)$ an $O(1)$ random variable at fixed depth.

2.7 Assumptions and order of limits

The calculations below assume:

1. fixed B, d_0 , and depth;
2. $n \rightarrow \infty$ through equal hidden widths;
3. enough smoothness of ϕ for every displayed Taylor and Stein derivative;

4. polynomial or otherwise controlled growth of the required activation derivatives;
5. uniform moment bounds for every retained contraction state;
6. leave-finitely-many-out control when a singled-out neuron also enters an empirical covariance;
7. uniform integrability when convergence in probability is converted into convergence of expectations;
8. joint conditional multivariate CLTs, including convergence of every mixed covariance state, for the finite collection of forward, backward, tangent, and fresh Gaussian base fields retained by each peel.

The Taylor coefficient is first defined at finite n . The width limit is then taken. Every later display using shorthand such as $o(\eta^2) + o_n(1)$ is to be read in this ordered, coefficientwise sense; no uniform two-parameter remainder estimate is claimed. Claims about a whole finite- η trajectory require additional uniform remainder estimates and are not implied merely by the coefficient calculations.

For ReLU, ordinary pointwise C^2 Taylor formulas are unavailable. A weak-derivative or Gaussian-boundary calculation is required.

3. The two exact Gaussian identities used in every peel

Let $w = (w_1, \dots, w_n)$ have independent $N(0, 1)$ coordinates and let

$$Z(a) = \frac{1}{\sqrt{n}} \sum_j w_j h_j(a).$$

Conditional on the lower-layer numbers $h_j(a)$, the vector (w, Z) is jointly Gaussian.

For a smooth function F of the fixed-batch vector Z , one-weight Stein gives

$$\mathbb{E}[w_p F(Z)] = \frac{1}{\sqrt{n}} \sum_a h_p(a) \mathbb{E}[\partial_a F(Z)]. \quad (3.1)$$

Applying Gaussian integration by parts twice gives

$$\mathbb{E}[w_p w_q F(Z)] = \delta_{pq} \mathbb{E}[F(Z)] + \frac{1}{n} \sum_{a,b} h_p(a) h_q(b) \mathbb{E}[\partial_a \partial_b F(Z)]. \quad (3.2)$$

The first term of (3.2) is the weight–weight Wick branch. The second is the Stein/Onsager branch. No independence between w and Z has been asserted.

For a top-row sum, equality patterns must be separated. For example, $p = q$ and $p \neq q$ have different numbers of free indices and different Gaussian contractions. A branch is discarded only after its complete index multiplicity and every lower-layer sum created by Stein differentiation have been counted.

4. Execution I: backward sensitivity kernels

Define

$$\Pi_{n,ab}^\ell = \frac{1}{n} \sum_i \delta_i^\ell(a) \delta_i^\ell(b).$$

We peel from the readout downward.

4.1 Readout and group 3

At the third hidden layer,

$$\Pi_{n,ab}^3 = \frac{1}{n} \sum_r v_r^2 \phi'(z_r^3(a)) \phi'(z_r^3(b)).$$

The readout weights are independent of all hidden preactivations, so

$$\mathbb{E}[v_r^2] = 1.$$

Conditional on the second hidden layer, the rows z_r^3 are iid Gaussian with covariance G_n^2 . A weighted law of large numbers followed by $G_n^2 \rightarrow G^2$ gives

$$\Pi_{n,ab}^3 \longrightarrow D_{ab}^3.$$

4.2 Full scalarization of the group-3 peel for Π^2

Substitute the recursion for δ^2 :

$$\begin{aligned} \Pi_{n,ab}^2 &= \frac{1}{n^2} \sum_{p,r,t} \phi'(z_p^2(a)) \phi'(z_p^2(b)) W_{rp}^3 W_{tp}^3 \\ &\quad \times v_r v_t \phi'(z_r^3(a)) \phi'(z_t^3(b)). \end{aligned} \quad (4.1)$$

The readout is the highest group. Peel it first:

$$\mathbb{E}_v[v_r v_t] = \delta_{rt}.$$

Thus

$$\mathbb{E}_v[\Pi_{n,ab}^2] = \frac{1}{n^2} \sum_{p,r} \phi'(z_p^2(a)) \phi'(z_p^2(b)) (W_{rp}^3)^2 \times \phi'(z_r^3(a)) \phi'(z_r^3(b)). \quad (4.2)$$

The factor $(W_{rp}^3)^2$ is not independent of z_r^3 . Apply (3.2) with $p = q$ and

$$F(Z) = \phi'(Z_a) \phi'(Z_b).$$

Because

$$\frac{\partial z_r^3(c)}{\partial W_{rp}^3} = \frac{h_p^2(c)}{\sqrt{n}},$$

the exact second derivative is

$$\begin{aligned} & \frac{\partial^2}{\partial (W_{rp}^3)^2} [\phi'(z_r^3(a)) \phi'(z_r^3(b))] \\ &= \frac{1}{n} \left[h_p^2(a)^2 \phi'''(z_r^3(a)) \phi'(z_r^3(b)) \right. \\ & \quad + 2h_p^2(a) h_p^2(b) \phi''(z_r^3(a)) \phi''(z_r^3(b)) \\ & \quad \left. + h_p^2(b)^2 \phi'(z_r^3(a)) \phi'''(z_r^3(b)) \right]. \end{aligned} \quad (4.3)$$

Let

$$\mathcal{E}_{n,ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G_n^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

After applying (3.2) and summing the group-3-only index r , the exact conditional expression is

$$\begin{aligned} & \mathbb{E}_{v, W^3} [\Pi_{n,ab}^2 \mid W^1, W^2] \\ &= \frac{1}{n} \sum_p \phi'(z_p^2(a)) \phi'(z_p^2(b)) \left[D_{n,ab}^3 + \frac{1}{n} \sum_{c,d} h_p^2(c) h_p^2(d) \mathcal{E}_{n,ab}^3(c, d) \right], \end{aligned} \quad (4.3a)$$

where

$$D_{n,ab}^3 = \mathbb{E}_{Z \sim N(0, G_n^2)} [\phi'(Z_a) \phi'(Z_b)].$$

Only now is the empirical covariance replaced by its deterministic limit. The first branch of (4.3a) converges to $D_{ab}^2 D_{ab}^3$. The second branch has the form $n^{-2} \sum_p$ of bounded-moment row terms and is $O(n^{-1})$. Therefore

$$\boxed{\mathbb{E}[\Pi_{n,ab}^2] \longrightarrow D_{ab}^2 D_{ab}^3.}$$

4.3 Full path scalarization for Π^1

Expanding both backward paths gives

$$\begin{aligned} \Pi_{n,ab}^1 &= \frac{1}{n^3} \sum_{i,p,q,r,t} v_r v_t W_{rp}^3 W_{rq}^3 W_{pi}^2 W_{qi}^2 \\ & \quad \times \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ & \quad \times \phi'(z_p^2(a)) \phi'(z_q^2(b)) \\ & \quad \times \phi'(z_r^3(a)) \phi'(z_t^3(b)). \end{aligned} \quad (4.4)$$

The index ownership is essential:

- after the readout peel, r is group-3-only;
- p, q occur in groups 3 and 2 and cannot be summed out during the group-3 peel;
- i occurs in groups 2 and 1 and cannot be summed out during the group-2 peel.

4.3.1 Peel the readout

Using

$$\mathbb{E}[v_r v_t] = \delta_{rt},$$

we obtain

$$\begin{aligned} \mathbb{E}_v[\Pi_{n,ab}^1] &= \frac{1}{n^3} \sum_{i,p,q,r} W_{rp}^3 W_{rq}^3 W_{pi}^2 W_{qi}^2 \\ & \quad \times \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ & \quad \times \phi'(z_p^2(a)) \phi'(z_q^2(b)) \\ & \quad \times \phi'(z_r^3(a)) \phi'(z_r^3(b)). \end{aligned} \quad (4.5)$$

4.3.2 Peel group 3

Condition on groups 1 and 2. Applying (3.2) to row r gives

$$\begin{aligned} & \mathbb{E}_{W^3} [W_{rp}^3 W_{rq}^3 \phi'(z_r^3(a)) \phi'(z_r^3(b))] \\ &= \delta_{pq} D_{n,ab}^3 + \frac{1}{n} \sum_{c,d} h_p^2(c) h_q^2(d) \mathcal{E}_{n,ab}^3(c, d), \end{aligned} \quad (4.6)$$

where

$$D_{n,ab}^3 = \mathbb{E}_{Z \sim N(0, G_n^2)} [\phi'(Z_a) \phi'(Z_b)],$$

and

$$\mathcal{E}_{n,ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G_n^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

The Stein-created factors $h_p^2(c) h_q^2(d)$ belong to group 2. They remain inside the lower boundary state.

The first branch of (4.6) enforces $p = q$. The second leaves both patterns $p = q$ and $p \neq q$ to be counted at group 2.

Before applying a group-2 Stein identity, remove the dependence of the coefficients in (4.6) on the distinguished group-2 rows. Using a leave-two-out version of G_n^2 and then its deterministic limit gives

$$D_{n,ab}^3 = D_{ab}^3 + o_{L^1}(1),$$

$$\mathcal{E}_{n,ab}^3(c, d) = \mathcal{E}_{ab}^3(c, d) + o_{L^1}(1), \quad (4.6a)$$

where

$$\mathcal{E}_{ab}^3(c, d) = \mathbb{E}_{Z \sim N(0, G^2)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}].$$

The $o(1)$ error is carried through the remaining normalized sums by uniform integrability. Equivalently, one could retain derivatives of D_n^3 and $\mathcal{E}_n^3$ with respect to the distinguished rows; the leave-two-out replacement is the cleaner leading-order implementation used here.

4.3.3 Peel group 2: Wick branch

The group-3 Wick branch becomes

$$\frac{D_{ab}^3}{n^2} \sum_{i,p} (W_{pi}^2)^2 \phi'(z_i^1(a)) \phi'(z_i^1(b)) \phi'(z_p^2(a)) \phi'(z_p^2(b)). \quad (4.7)$$

Applying (3.2) to $(W_{pi}^2)^2$ gives a leading Wick term

$$\frac{D_{ab}^3 D_{n,ab}^2}{n} \sum_i \phi'(z_i^1(a)) \phi'(z_i^1(b)), \quad (4.8)$$

up to the already recorded $o(1)$ covariance-replacement error.

More explicitly, if

$$\mathcal{E}_{n,ab}^2(c, d) = \mathbb{E}_{Z \sim N(0, G_n^1)} [\partial_c \partial_d \{\phi'(Z_a) \phi'(Z_b)\}],$$

then

$$\begin{aligned} & \mathbb{E}_{W^2} [(W_{pi}^2)^2 \phi'(z_p^2(a)) \phi'(z_p^2(b))] \\ &= D_{n,ab}^2 + \frac{1}{n} \sum_{c,d} h_i^1(c) h_i^1(d) \mathcal{E}_{n,ab}^2(c, d). \end{aligned} \quad (4.8a)$$

After the p - and i -sums, the second line contributes $O(n^{-1})$ and leaves only group-1 activations.

4.3.4 Peel group 2: branch created by group-3 Stein differentiation

After the group-3-only index r has been summed, the full branch created by the second term of (4.6) is

$$\begin{aligned} \mathcal{S}_n &= \frac{1}{n^3} \sum_{i,p,q} W_{pi}^2 W_{qi}^2 \phi'(z_i^1(a)) \phi'(z_i^1(b)) \\ &\quad \times \sum_{c,d} \mathcal{E}_{ab}^3(c, d) \phi'(z_p^2(a)) \phi(z_p^2(c)) \\ &\quad \times \phi'(z_q^2(b)) \phi(z_q^2(d)). \end{aligned} \quad (4.9)$$

For $p = q$, define

$$P_{cd}(Z) = \phi'(Z_a) \phi'(Z_b) \phi(Z_c) \phi(Z_d).$$

Conditional on group 1, let $Z \sim N(0, G_n^1)$. The exact group-2 row peel of the row-dependent factor is

$$\begin{aligned} & \mathbb{E}[(W_{pi}^2)^2 P_{cd}(z_p^2)] \\ &= \mathbb{E}[P_{cd}(Z)] + \frac{1}{n} \sum_{u,w} h_i^1(u) h_i^1(w) \mathbb{E}[\partial_u \partial_w P_{cd}(Z)]. \end{aligned} \quad (4.10)$$

The diagonal family contains only n^2 choices for (i, p) . With the n^{-3} normalization in (4.9), its Wick and Stein terms are respectively $O(n^{-1})$ and $O(n^{-2})$.

For $p \neq q$, the two group-2 rows are conditionally independent. Each single weight must be removed using the one-weight identity (3.1). The two Stein covariances contribute n^{-1} together. There are $n^3(1 + o(1))$ choices for (i, p, q) , but the original normalization is n^{-3} , so this branch is again $O(n^{-1})$.

In full notation, put

$$F_{a,c}(Z) = \phi'(Z_a)\phi(Z_c), \quad F_{b,d}(Z) = \phi'(Z_b)\phi(Z_d).$$

For $p \neq q$,

$$\begin{aligned} \mathbb{E}[W_{pi}^2 F_{a,c}(z_p^2)] &= \frac{1}{\sqrt{n}} \sum_u h_i^1(u) \mathbb{E}[\partial_u F_{a,c}(Z)], \\ \mathbb{E}[W_{qi}^2 F_{b,d}(z_q^2)] &= \frac{1}{\sqrt{n}} \sum_w h_i^1(w) \mathbb{E}[\partial_w F_{b,d}(Z)]. \end{aligned} \quad (4.11)$$

Their product supplies the stated n^{-1} Stein cost. Thus the five branch orders—leading ladder, group-2 Stein correction, group-3-Stein diagonal Wick, group-3-Stein diagonal Stein, and group-3-Stein off-diagonal—are

$$1, \quad n^{-1}, \quad n^{-1}, \quad n^{-2}, \quad n^{-1}. \quad (4.12)$$

Thus every branch created by the group-3 Stein term vanishes at leading order, but only after the $p = q$ and $p \neq q$ families have been separated and counted.

Before the final group-1 law of large numbers, use a leave-one-out covariance to replace

$$D_{n,ab}^2 = D_{ab}^2 + o_{L^1}(1), \quad \mathcal{C}_{n,ab}^2(c, d) = \mathcal{C}_{ab}^2(c, d) + o_{L^1}(1). \quad (4.12a)$$

This prevents the distinguished group-1 row from being hidden inside a coefficient that is incorrectly treated as constant.

4.3.5 Peel group 1

After group 2 is removed, the leading remaining expression is (4.8). The index i now belongs only to group 1, so the law of large numbers gives

$$\frac{1}{n} \sum_i \phi'(z_i^1(a)) \phi'(z_i^1(b)) \longrightarrow D_{ab}^1.$$

Therefore

$$\mathbb{E}[\Pi_{n,ab}^1] \longrightarrow D_{ab}^1 D_{ab}^2 D_{ab}^3.$$

4.4 Result and concentration obligation

The limiting backward recursion is

$$\Pi^3 = D^3, \quad \Pi^2 = D^2 \odot \Pi^3, \quad \Pi^1 = D^1 \odot \Pi^2.$$

Here $\odot$ denotes entrywise multiplication.

The calculation above establishes the expectation limits. To obtain convergence of the random empirical kernel itself, one must peel the square. Four backward paths appear. The expected mechanism is that the disconnected pair of ladders reproduces the square of the expectation, while every connected equality pattern loses at least one free neuron index. A sufficient estimate is

$$\text{Var}(\Pi_{n,ab}^\ell) = O(n^{-1}).$$

The complete four-path Wick–Stein enumeration is not reproduced in this case study. Accordingly, the variance estimate is recorded as an additional concentration hypothesis/proof obligation under the fixed-depth moment assumptions. When it holds,

$$\Pi_{n,ab}^\ell \rightarrow \Pi_{ab}^\ell$$

in L^2 . Every later replacement of a product involving Π_n^ℓ invokes this concentration estimate together with uniform integrability; it does not follow from the one-copy expectation calculation alone.

4.5 Exact conditional integration versus mean-field replacement

At group 3 the Gaussian covariance used in (4.3a) and (4.6) is initially

$$G_{n,cd}^2 = \frac{1}{n} \sum_p h_p^2(c) h_p^2(d),$$

not the deterministic matrix G^2 . Likewise, the group-2 row peel initially uses

$$G_{n,cd}^1 = \frac{1}{n} \sum_i h_i^1(c) h_i^1(d).$$

The correct order is:

1. condition on the lower groups and apply the exact Gaussian integration-by-parts identity with $G_n^{\ell-1}$;
2. retain every lower-group factor created by differentiation;
3. expose the equality patterns of the distinguished rows;
4. then replace $G_n^{\ell-1}$ by $G^{\ell-1}$.

When a distinguished row p or pair (p, q) also occurs inside G_n^2 , a quantitative proof replaces G_n^2 temporarily by its leave-one-out or leave-two-out version. Removing finitely many rows changes the covariance by $O(n^{-1})$; the remaining covariance is independent of the distinguished rows and converges to G^2 . Uniform integrability then permits passage to expectations.

This procedure proves the leading $o(1)$ limit. It does not, without a sharper fluctuation expansion, identify an exact finite-width $1/n$ bias.

5. Execution II: one gradient-descent step

The output and sample losses are computed through linear order in η . The hidden Grams are computed through quadratic order, because their deterministic linear coefficient vanishes. All Taylor coefficients are formed at finite width before taking $n \rightarrow \infty$.

5.1 Exact parameter and preactivation velocities

Put a dot over a quantity to mean its derivative with respect to the one-step learning-rate parameter at $\eta = 0$. From

$$\theta^+ = \theta - n\eta \nabla_{\theta} \mathcal{J}$$

and the definition of δ , the exact raw-parameter velocities are

$$\dot{v}_r = -\frac{2}{B} \sum_c e_c h_r^3(c), \quad (5.1)$$

$$\dot{W}_{rp}^3 = -\frac{2}{B\sqrt{n}} \sum_c e_c \delta_r^3(c) h_p^2(c), \quad (5.2)$$

$$\dot{W}_{pi}^2 = -\frac{2}{B\sqrt{n}} \sum_c e_c \delta_p^2(c) h_i^1(c), \quad (5.3)$$

and

$$\dot{W}_{ij}^1 = -\frac{2}{B\sqrt{d_0}} \sum_c e_c \delta_i^1(c) x_{cj}. \quad (5.4)$$

Define the first preactivation velocities

$$u_i^\ell(a) := \left. \frac{dz_i^{\ell,+}(a)}{d\eta} \right|_{\eta=0}.$$

Substitution of (5.2)–(5.4) into the forward equations gives, still exactly at finite width,

$$u_i^1(a) = -\frac{2}{B} \sum_c e_c G_{ac}^0 \delta_i^1(c), \quad (5.5)$$

$$\begin{aligned} u_p^2(a) = & -\frac{2}{B} \sum_c e_c G_{n,ac}^1 \delta_p^2(c) \\ & + \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 \phi'(z_i^1(a)) u_i^1(a), \end{aligned} \quad (5.6)$$

and

$$\begin{aligned} u_r^3(a) = & -\frac{2}{B} \sum_c e_c G_{n,ac}^2 \delta_r^3(c) \\ & + \frac{1}{\sqrt{n}} \sum_p W_{rp}^3 \phi'(z_p^2(a)) u_p^2(a). \end{aligned} \quad (5.7)$$

These equations already exhibit the two branches that peeling must track: the direct change of the current matrix and the propagated change of the preceding representation.

5.2 Exact tangent kernel and the linear loss coefficient

Define the optimizer-metric tangent kernel

$$K_{n,ab} = n \sum_{k=1}^3 \langle \nabla_{W^k} f_a, \nabla_{W^k} f_b \rangle + n \langle \nabla_v f_a, \nabla_v f_b \rangle. \quad (5.8)$$

The finite-width output velocity is therefore

$$\dot{f}_a = -\frac{2}{B} \sum_c K_{n,ac} e_c. \quad (5.9)$$

Every parameter gradient is rank one. For example,

$$\frac{\partial f_a}{\partial W_{rp}^3} = \frac{\delta_r^3(a) h_p^2(a)}{n\sqrt{n}},$$

so

$$\begin{aligned} n \sum_{r,p} \frac{\partial f_a}{\partial W_{rp}^3} \frac{\partial f_b}{\partial W_{rp}^3} &= \left(\frac{1}{n} \sum_r \delta_r^3(a) \delta_r^3(b) \right) \left(\frac{1}{n} \sum_p h_p^2(a) h_p^2(b) \right) \\ &= \Pi_{n,ab}^3 G_{n,ab}^2. \end{aligned}$$

Doing the same for every block gives the exact factorization

$$K_{n,ab} = G_{n,ab}^3 + G_{n,ab}^2 \Pi_{n,ab}^3 + G_{n,ab}^1 \Pi_{n,ab}^2 + G_{n,ab}^0 \Pi_{n,ab}^1. \quad (5.10)$$

The backward expectation calculation in Section 4, together with the joint concentration obligation stated in Section 4.4, gives

$$\begin{aligned} K_{ab} &= G_{ab}^3 + G_{ab}^2 D_{ab}^3 + G_{ab}^1 D_{ab}^2 D_{ab}^3 \\ &\quad + G_{ab}^0 D_{ab}^1 D_{ab}^2 D_{ab}^3. \end{aligned} \quad (5.11)$$

Because $f_a \rightarrow 0$, the residual satisfies $e_a \rightarrow -y_a$. Weighted convergence, rather than a finite-width independence claim, yields

$$\lim_{n \rightarrow \infty} \mathbb{E}[\dot{f}_a] = \frac{2}{B} \sum_c K_{ac} y_c. \quad (5.12)$$

For the individual loss $\ell_a = e_a^2$,

$$\dot{\ell}_a = 2e_a \dot{f}_a = -\frac{4}{B} e_a \sum_c K_{n,ac} e_c.$$

Consequently,

$$\mathbb{E}[\ell_a^+] = y_a^2 - \frac{4\eta}{B} y_a \sum_c K_{ac} y_c + o(\eta) + o_n(1). \quad (5.13)$$

For the average loss this becomes

$$\mathbb{E}[\mathcal{J}^+] = \frac{1}{B} \sum_a y_a^2 - \frac{4\eta}{B^2} y^\top K y + o(\eta) + o_n(1). \quad (5.14)$$

The batch-average coefficient is nonpositive because K is positive semidefinite. A particular sample loss need not decrease.

5.3 Exact hidden-Gram linear coefficient

Define the activation velocity

$$g_i^\ell(a) := \left. \frac{dh_i^{\ell,+}(a)}{d\eta} \right|_0 = \phi'(z_i^\ell(a)) u_i^\ell(a). \quad (5.15)$$

The coefficient of η in the hidden Gram is exactly

$$L_{n,ab}^\ell := \left. \frac{dG_{n,ab}^{\ell,+}}{d\eta} \right|_0 = \frac{1}{n} \sum_i [g_i^\ell(a) h_i^\ell(b) + h_i^\ell(a) g_i^\ell(b)]. \quad (5.16)$$

Equivalently, before any width limit,

$$L_{n,ab}^\ell = -\frac{2n}{B} \sum_c e_c \sum_{k \leq \ell} \left\langle \nabla_{W^k} G_{n,ab}^\ell, \nabla_{W^k} f_c \right\rangle. \quad (5.17)$$

The readout must be peeled before any hidden group. For a coordinate $z_j^k(c)$ with $k \leq 3$,

$$\delta_j^k(c) = \sum_r v_r \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)}$$

is linear in v . Hence the label branch is odd and vanishes:

$$\mathbb{E}_v[-y_c \delta_j^k(c)] = 0. \quad (5.18)$$

The output branch contains two readout factors. Its contraction is

$$\begin{aligned} \mathbb{E}_v[f_c \delta_j^k(c)] &= \frac{1}{n} \sum_{r,t} \mathbb{E}[v_t v_r] h_t^3(c) \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)} \\ &= \frac{1}{n} \sum_r h_r^3(c) \phi'(z_r^3(c)) \frac{\partial z_r^3(c)}{\partial z_j^k(c)} \\ &= \frac{1}{2} \frac{\partial G_{n,cc}^3}{\partial z_j^k(c)}. \end{aligned} \quad (5.19)$$

Substituting this exact top-group peel into (5.17) gives

$$\mathbb{E}_v[L_{n,ab}^\ell \mid W^1, W^2, W^3] = -\frac{1}{B} \sum_c \sum_{k \leq \ell} \left\langle \nabla_{W^k} G_{n,ab}^\ell, \nabla_{W^k} G_{n,cc}^3 \right\rangle. \quad (5.20)$$

This formula makes the remaining width loss visible. For example, the direct $k = \ell = 3$ term is

$$\begin{aligned} &-\frac{2}{Bn} \sum_c \left[G_{n,ac}^2 \frac{1}{n} \sum_r h_r^3(b) \phi'(z_r^3(a)) h_r^3(c) \phi'(z_r^3(c)) \right. \\ &\quad \left. + G_{n,bc}^2 \frac{1}{n} \sum_r h_r^3(a) \phi'(z_r^3(b)) h_r^3(c) \phi'(z_r^3(c)) \right]. \end{aligned} \quad (5.21)$$

The two inner averages have finite Gaussian limits, while the outer $1/n$ remains. For $k < \ell$, expanding the two derivatives in (5.20) creates two forward paths. Peeling from group 3 downward gives either a Wick pairing or a Stein attachment at each shared matrix. Those operations change the limiting Gaussian moment but do not restore the one free neuron index already lost in (5.20). Thus every such path family is also $O(n^{-1})$.

Therefore

$$\mathbb{E}[L_{n,ab}^\ell] = O(n^{-1}),$$

and

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[G_{n,ab}^{\ell,+}] = 0. \quad (5.22)$$

The cancellation is an annealed statement caused by the centered readout. It does not say that an individual finite-width network has zero first-order feature motion.

5.4 Exact quadratic hidden-Gram expansion

At finite width write

$$z_i^{\ell,+}(a) = z_i^\ell(a) + \eta u_i^\ell(a) + \eta^2 q_i^\ell(a) + o(\eta^2), \quad (5.23)$$

and

$$h_i^{\ell,+}(a) = h_i^\ell(a) + \eta g_i^\ell(a) + \eta^2 s_i^\ell(a) + o(\eta^2). \quad (5.24)$$

Taylor's formula gives

$$g_i^\ell(a) = \phi'(z_i^\ell(a)) u_i^\ell(a), \quad (5.25)$$

$$s_i^\ell(a) = \phi'(z_i^\ell(a)) q_i^\ell(a) + \frac{1}{2} \phi''(z_i^\ell(a)) u_i^\ell(a)^2. \quad (5.26)$$

The first preactivation is linear in W^1 , so

$$q^1 = 0.$$

For $\ell = 2, 3$, expanding the product of the updated matrix and the updated preceding activation gives

$$q_i^\ell(a) = \frac{1}{\sqrt{n}} \sum_j [W_{ij}^\ell s_j^{\ell-1}(a) + \dot{W}_{ij}^\ell g_j^{\ell-1}(a)]. \quad (5.27)$$

Therefore the exact coefficient of η^2 in the hidden Gram is

$$C_{n,ab}^\ell = \frac{1}{n} \sum_i \left[g_i^\ell(a) g_i^\ell(b) + s_i^\ell(a) h_i^\ell(b) + h_i^\ell(a) s_i^\ell(b) \right]. \quad (5.28)$$

5.5 Peel the residual and readout factors in the quadratic coefficient

Every first velocity is linear in the residuals. Its leading quadratic products therefore contain

$$e_c e_d = y_c y_d - y_c f_d - y_d f_c + f_c f_d. \quad (5.29)$$

Each normalized backward signal δ is odd under $v \mapsto -v$. A quadratic velocity term has two such signals and is even. The two mixed terms in (5.29) add one further readout factor and are odd, so their conditional readout expectation is zero. The $f_c f_d$ branch is even but is $O(n^{-1})$ in every fixed moment because $f_c = O_{L^p}(n^{-1/2})$. Thus only the label-label term survives at leading order.

Set

$$\lambda_c := \frac{2y_c}{B}. \quad (5.30)$$

From this point through the end of the one-step quadratic calculation, all velocities mean their leading label-driven branch.

5.6 Separate the local activation gate from the incoming backward signal

At finite width define the incoming backward signals by

$$\begin{aligned} \xi_r^3(a) &= v_r, \\ \xi_p^2(a) &= \frac{1}{\sqrt{n}} \sum_r W_{rp}^3 \delta_r^3(a), \\ \xi_i^1(a) &= \frac{1}{\sqrt{n}} \sum_p W_{pi}^2 \delta_p^2(a). \end{aligned} \quad (5.31a)$$

Then the exact identity is

$$\delta_i^\ell(a) = \phi'(z_i^\ell(a)) \xi_i^\ell(a).$$

For a limiting representative neuron, write

$$\delta^\ell(a) = \phi'(Z_a^\ell) \xi^\ell(a). \quad (5.31)$$

The multichannel extension of the Section 4 peel gives, under the joint-CLT and leave-one-out assumptions in Section 2.7, Gaussian base fields satisfying

$$Z^\ell \perp \xi^\ell,$$

with

$$\mathbb{E}[\xi^\ell(a) \xi^\ell(b)] = P_{ab}^\ell,$$

where

$$P_{ab}^3 = 1, \quad P_{ab}^2 = D_{ab}^3, \quad P_{ab}^1 = D_{ab}^2 D_{ab}^3. \quad (5.32)$$

Accordingly,

$$\Pi_{ab}^\ell = D_{ab}^\ell P_{ab}^\ell.$$

The displayed independence is an additional leading joint-state conclusion, not a consequence of the backward second moment alone. It follows by conditioning on the forward network, peeling the mixed moments and conditional covariance of ξ^ℓ , removing finitely many distinguished rows, and applying the conditional multivariate CLT. It is not a finite-width claim that a backward signal is independent of the weights and preactivations that define it.

Define three partial tangent factors, entrywise in the pair (a, b) :

$$T^0 = G^0,$$

$$T^1 = G^1 + D^1 \odot T^0,$$

$$T^2 = G^2 + D^2 \odot T^1. \quad (5.33)$$

Thus $T^{\ell-1}$ records the parameters at and below layer ℓ that can move z^ℓ .

5.7 Execute the first-velocity peel

At layer 1, (5.5) immediately becomes

$$u_i^1(a) = \sum_c \lambda_c T_{ac}^0 \delta_i^1(c). \quad (5.34)$$

At layer 2, substitute (5.34) and the explicit transpose channel

$$\xi_i^1(c) = \frac{1}{\sqrt{n}} \sum_{p'} W_{p'i}^2 \delta_{p'}^2(c)$$

into the propagated term of (5.6). This exposes all indices:

$$\begin{aligned} & \frac{1}{\sqrt{n}} \sum_i W_{pi}^2 \phi'(z_i^1(a)) u_i^1(a) \\ &= \frac{1}{n} \sum_{i,p'} W_{pi}^2 W_{p'i}^2 \phi'(z_i^1(a)) \sum_c \lambda_c G_{ac}^0 \phi'(z_i^1(c)) \delta_{p'}^2(c). \end{aligned} \quad (5.35)$$

The equality patterns $p' = p$ and $p' \neq p$ must be treated separately.

For $p' = p$, the Wick branch of the two-weight identity forces the two paths to use the same target row. After the group-1 average is exposed, it gives

$$\sum_c \lambda_c G_{ac}^0 D_{ac}^1 \delta_p^2(c). \quad (5.36)$$

Adding the direct matrix-update term in (5.6),

$$\sum_c \lambda_c G_{ac}^1 \delta_p^2(c),$$

produces

$$\sum_c \lambda_c T_{ac}^1 \delta_p^2(c). \quad (5.37)$$

For $p' \neq p$, an individual term has zero centered conditional mean, but the whole family cannot be discarded: it contains $n^2(1+o(1))$ pairs (i, p') . Scalarizing two copies of (5.35), pairing the row-distinct weights, and then applying the group-1 law of large numbers leaves a centered Gaussian field. Call it Γ^2 . Its covariance is exactly the preceding activation-velocity covariance,

$$\mathbb{E}[\Gamma^2(a) \Gamma^2(b)] = \mathbb{E}[g^1(a) g^1(b)]. \quad (5.38)$$

The same diagonal-response/off-diagonal-fluctuation split at layer 3 yields the following equality in the limiting representative-neuron law:

$$\boxed{u^\ell(a) = \Gamma^\ell(a) + \sum_c \lambda_c T_{ac}^{\ell-1} \delta^\ell(c), \quad \ell = 1, 2, 3,} \quad (5.39)$$

where $\Gamma^1 = 0$. For $\ell = 2, 3$, the fresh field is centered Gaussian and

$$\mathbb{E}[\Gamma^\ell(a) \Gamma^\ell(b)] = V_{ab}^{\ell-1}, \quad (5.40)$$

where

$$V_{ab}^\ell := \mathbb{E}[g^\ell(a) g^\ell(b)]. \quad (5.41)$$

The remaining cross-covariance with the current preactivation is

$$\mathbb{E}[\Gamma^\ell(a) Z_b^\ell] = \mathbb{E}[g^{\ell-1}(a) h^{\ell-1}(b)].$$

The right-hand side is odd in the centered readout and is zero at this order. The recursion below additionally uses the corresponding mixed two-copy conclusion

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[\left(\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell g_j^{\ell-1}(a) - \text{its complete Stein response} \right) \xi_i^\ell(b) \right] = 0. \quad (5.40a)$$

Its disconnected branch vanishes by readout parity; every connected branch must reconnect the two copies through a shared row or a Stein attachment and loses one width power. A complete joint-state theorem must include this mixed contraction explicitly. Under that joint-CLT obligation, zero cross-covariance of the Gaussian base fields makes Γ^ℓ independent of (Z^ℓ, ξ^ℓ) at this one-step order.

5.8 Recursion for the velocity covariance

Insert (5.31) and (5.39) into

$$g^\ell(a) = \phi'(Z_a^\ell) u^\ell(a).$$

The fresh-field/response cross terms vanish. The fresh-field square gives $D_{ab}^\ell V_{ab}^{\ell-1}$. The response square gives one backward covariance P_{cd}^ℓ and a four-gate Gaussian moment. Therefore

$$\begin{aligned}
V_{ab}^\ell &= D_{ab}^\ell V_{ab}^{\ell-1} \\
&\quad + \sum_{c,d} \lambda_c \lambda_d T_{ac}^{\ell-1} T_{bd}^{\ell-1} P_{cd}^\ell \\
&\quad \times \mathbb{E}[\phi'(Z_a^\ell) \phi'(Z_b^\ell) \phi'(Z_c^\ell) \phi'(Z_d^\ell)],
\end{aligned} \tag{5.42}$$

with $V^0 = 0$ and $Z^\ell \sim N(0, G^{\ell-1})$.

The first line of (5.42) is the off-diagonal Wick/CLT branch just derived from $p' \neq p$. Replacing the empirical random operator in (5.35) by its mean before squaring would delete this term and give a wrong quadratic coefficient.

5.9 Execute the second-displacement peel

Define the cross moment

$$S_{ab}^\ell := \mathbb{E}[s^\ell(a) h^\ell(b)], \quad S^0 = 0. \tag{5.43}$$

There are two terms in (5.27). The mixed matrix/feature term is, on the label branch,

$$\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell g_j^{\ell-1}(a) = \sum_c \lambda_c \delta_i^\ell(c) X_{n,ca}^{\ell-1}, \tag{5.44}$$

where

$$X_{n,ca}^{\ell-1} = \frac{1}{n} \sum_j h_j^{\ell-1}(c) g_j^{\ell-1}(a). \tag{5.45}$$

Its one-copy expectation is zero by readout parity, but that is not sufficient to discard it. Scalarizing two copies gives

$$\mathbb{E}[(X_{n,ca}^{\ell-1})^2] = \frac{1}{n^2} \sum_{j,k} \mathbb{E}[h_j(c) g_j(a) h_k(c) g_k(a)]. \tag{5.46}$$

The $j = k$ family has n assignments against the n^2 normalization and is $O(n^{-1})$. In the $j \neq k$ family, the disconnected readout contraction factorizes into two zero odd first moments. A connected contraction must either identify a current-layer row or attach one copy to the other by a Stein covariance carrying an extra n^{-1} . The lower-group boundary sums created by that attachment cannot restore the lost width: both copies must reconnect to the same current-layer backward field. Thus the connected family is also $O(n^{-1})$. Hence

$$X_{n,ca}^{\ell-1} = O_{L^2}(n^{-1/2}),$$

and (5.44) is negligible in the leading annealed coefficient after bounded-moment control.

It remains to peel

$$\frac{1}{\sqrt{n}} \sum_j W_{ij}^\ell s_j^{\ell-1}(a). \tag{5.47}$$

The target-row Stein response coefficient is

$$\mathbb{E} \left[\frac{\partial s^{\ell-1}(a)}{\partial \xi^{\ell-1}(c)} \right].$$

At this order $s^{\ell-1}$ is even under $v \mapsto -v$, whereas $\xi^{\ell-1}$ is odd. The differentiated random quantity is therefore odd, so its annealed response coefficient—not the random derivative itself—is zero. In the limiting representative-neuron law, the row-distinct Wick patterns leave a centered Gaussian base field Ω^ℓ with the covariance needed below:

$$\mathbb{E}[\Omega^\ell(a) Z_b^\ell] = S_{ab}^{\ell-1}. \tag{5.48}$$

Now use (5.26). Gaussian integration by parts in the jointly Gaussian pair (Ω^ℓ, Z^ℓ) gives

$$\begin{aligned}
&\mathbb{E}[\phi(Z_b^\ell) \phi'(Z_a^\ell) \Omega^\ell(a)] \\
&= D_{ab}^\ell S_{ab}^{\ell-1} + \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell)] S_{aa}^{\ell-1}.
\end{aligned} \tag{5.49}$$

The term $\frac{1}{2} \phi''(Z_a^\ell) u^\ell(a)^2$ has a fresh-field square and a response square. Combining them with (5.49) yields

$$\begin{aligned}
S_{ab}^\ell &= D_{ab}^\ell S_{ab}^{\ell-1} \\
&\quad + \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell)] \left(S_{aa}^{\ell-1} + \frac{1}{2} V_{aa}^{\ell-1} \right) \\
&\quad + \frac{1}{2} \sum_{c,d} \lambda_c \lambda_d T_{ac}^{\ell-1} T_{ad}^{\ell-1} P_{cd}^\ell \\
&\quad \times \mathbb{E}[\phi(Z_b^\ell) \phi''(Z_a^\ell) \phi'(Z_c^\ell) \phi'(Z_d^\ell)].
\end{aligned} \tag{5.50}$$

Every expectation in (5.42) and (5.50) is over the single B -dimensional vector

$$Z^\ell \sim N(0, G^{\ell-1}).$$

Finally, (5.28) becomes

$$C_{ab}^\ell = V_{ab}^\ell + S_{ab}^\ell + S_{ba}^\ell. \quad (5.51)$$

Thus the one-step hidden-Gram expansion is

$$\mathbb{E}[G_{n,ab}^{\ell,+}] = G_{ab}^\ell + \eta O(n^{-1}) + \eta^2 C_{ab}^\ell + o(\eta^2) + o_n(1). \quad (5.52)$$

5.10 Fully explicit first-layer coefficient

At layer 1, $V^0 = S^0 = 0$, $T^0 = G^0$, and $P_{cd}^1 = D_{cd}^2 D_{cd}^3$. Writing, only in the next display,

$$\phi_a = \phi(Z_a^1), \quad \phi'_a = \phi'(Z_a^1), \quad \phi''_a = \phi''(Z_a^1),$$

with $Z^1 \sim N(0, G^0)$, gives

$$C_{ab}^1 = \frac{4}{B^2} \sum_{cd} y_c y_d P_{cd}^1 \left[G_{ac}^0 G_{bd}^0 \mathbb{E}[\phi'_a \phi'_b \phi'_c \phi'_d] + \frac{1}{2} G_{ac}^0 G_{ad}^0 \mathbb{E}[\phi_b \phi''_a \phi'_c \phi'_d] + \frac{1}{2} G_{bc}^0 G_{bd}^0 \mathbb{E}[\phi_a \phi''_b \phi'_c \phi'_d] \right]. \quad (5.53)$$

This is already a completely explicit fixed-dimensional Gaussian integral. Layers 2 and 3 are obtained by evaluating (5.42), (5.50), and (5.51) in order.

5.11 Deep-linear audit of the one-step coefficient

Set $\phi(z) = z$. Then $D^\ell = P^\ell = 1$, $S^\ell = 0$, and every $G^\ell = G^0$. Define

$$\kappa = \frac{2}{B}, \quad \sigma_a = \sum_c G_{ac}^0 y_c.$$

The partial tangents are

$$T^0 = G^0, \quad T^1 = 2G^0, \quad T^2 = 3G^0.$$

The velocity recursion gives

$$C_{ab}^1 = \kappa^2 \sigma_a \sigma_b, \quad C_{ab}^2 = 5\kappa^2 \sigma_a \sigma_b, \quad C_{ab}^3 = 14\kappa^2 \sigma_a \sigma_b. \quad (5.54)$$

The nontrivial coefficient 5 can be checked directly. Put

$$M = \frac{1}{n} W^2 W^{2\top}.$$

The layer-2 velocity contains $I + M$, so

$$\frac{1}{n} \text{tr}(I + M)^2 \longrightarrow 1 + 2 \cdot 1 + 2 = 5,$$

because

$$\frac{1}{n} \text{tr} M \rightarrow 1, \quad \frac{1}{n} \text{tr} M^2 \rightarrow 2.$$

Replacing M by its entrywise mean I before forming the square would give 4, not 5. This solvable case directly verifies that the row-distinct fluctuation branch in (5.42) is essential.

6. Execution III: two gradient-descent steps

The second Euler step differs from simply doubling the first displacement because the gradient is recomputed at the parameters reached after step one. This section isolates that correction exactly, and gives the layer-by-layer state recursion needed to peel the hidden groups.

6.1 Exact two-step Taylor identity

Let the raw-coordinate μP vector field be

$$F(\theta) := -n \nabla \mathcal{J}(\theta). \quad (6.1)$$

Two Euler steps are

$$\theta^{(1)} = \theta + \eta F(\theta),$$

$$\theta^{(2)} = \theta^{(1)} + \eta F(\theta^{(1)}).$$

Taylor-expanding the second vector-field evaluation at finite width gives

$$\theta^{(2)} = \theta + 2\eta F + \eta^2 DF[F] + O(\eta^3). \quad (6.2)$$

For the observable

$$O(\theta) = G_{n,ab}^\ell(\theta),$$

define

$$L_{n,ab}^\ell = DO[F], \quad C_{n,ab}^\ell = \frac{1}{2} D^2 O[F, F], \quad R_{n,ab}^\ell = DO[DF[F]]. \quad (6.3)$$

Then

$$G_{n,ab}^{\ell,(2)} = G_{n,ab}^\ell + 2\eta L_{n,ab}^\ell + \eta^2 (4C_{n,ab}^\ell + R_{n,ab}^\ell) + O(\eta^3). \quad (6.4)$$

Subtracting the one-step expansion gives the change caused by the second update alone:

$$G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)} = \eta L_{n,ab}^\ell + \eta^2 (3C_{n,ab}^\ell + R_{n,ab}^\ell) + O(\eta^3). \quad (6.5)$$

The factors 4 and 3 are purely Taylor-combinatorial. The identity

$$D_F L_{n,ab}^\ell = 2C_{n,ab}^\ell + R_{n,ab}^\ell \quad (6.6)$$

is an independent check.

Section 5 proved $\mathbb{E}[L_{n,ab}^\ell] = O(n^{-1})$. Hence neither (6.4) nor (6.5) has a deterministic linear term after the sequential Taylor-then-width limit. Crucially, one may not set L_n^ℓ equal to its limiting value zero before differentiating it: (6.6) generally has a nonzero limit.

6.2 Isolate the new gradient correction

For squared loss,

$$\nabla^2 \mathcal{J} = \frac{2}{B} \sum_c \left(\nabla f_c \nabla f_c^\top + e_c \nabla^2 f_c \right). \quad (6.7)$$

Since $DF[F] = n^2 \nabla^2 \mathcal{J} \nabla \mathcal{J}$, substitution into (6.3) yields the exact finite-width identity

$$R_{n,ab}^\ell = \frac{4}{B^2} \sum_{c,d} e_d \left[\left(n \nabla G_{n,ab}^\ell \right)^\top \nabla f_c K_{n,cd} + e_c n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \right]. \quad (6.8)$$

The first line is the residual-change, or Gauss–Newton, branch. Its label part is odd in the centered readout and vanishes. Its surviving initialization–output branch is the same connected one-free-index-deficient family encountered in the linear Gram calculation, so it is $O(n^{-1})$.

In the second line, the label–label branch is even and survives, while mixed label/output branches are odd and the output–output branch is width-suppressed. Thus the leading correction is

$$R_{ab}^\ell = \frac{4}{B^2} \sum_{c,d} y_c y_d \lim_{n \rightarrow \infty} \mathbb{E} \left[n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \right]. \quad (6.9)$$

6.3 Execute the top readout peel exactly

Let θ and ψ denote individual raw parameter coordinates. Because a hidden Gram has no direct dependence on v , the contraction inside (6.9) is

$$n^2 \sum_{\theta \text{ hidden}} \sum_{\psi} \partial_{\theta} G_{n,ab}^\ell \partial_{\theta\psi} f_c \partial_{\psi} f_d. \quad (6.10)$$

There are two equality types for ψ .

If $\psi = v_s$, then

$$\partial_{\theta v_s} f_c = \frac{1}{n} \partial_{\theta} h_s^3(c), \quad \partial_{v_s} f_d = \frac{1}{n} h_s^3(d),$$

and the prefactor n^2 cancels the two readout normalizations.

If ψ is hidden, then

$$\partial_{\theta\psi} f_c = \frac{1}{n} \sum_r v_r \partial_{\theta\psi} h_r^3(c),$$

$$\partial_\psi f_d = \frac{1}{n} \sum_t v_t \partial_\psi h_t^3(d).$$

The exact readout Wick contraction $\mathbb{E}[v_r v_t] = \delta_{rt}$ again cancels the two factors $1/n$. Therefore

$$\begin{aligned} & \mathbb{E}_v \left[n^2 \left(\nabla G_{n,ab}^\ell \right)^\top \nabla^2 f_c \nabla f_d \middle| W^1, W^2, W^3 \right] \\ &= \sum_{\theta \text{ hidden}} \partial_\theta G_{n,ab}^\ell \left[\sum_r h_r^3(d) \partial_\theta h_r^3(c) \right. \\ & \quad \left. + \sum_{\psi \text{ hidden}} \sum_r \partial_{\theta\psi} h_r^3(c) \partial_\psi h_r^3(d) \right]. \end{aligned} \quad (6.11)$$

Equation (6.11) is the completely explicit boundary state handed from the readout group to group 3. No independence between a hidden matrix and the preactivations it creates has been used.

6.4 Exact differentiated forward and backward states

The Hessian contraction can be evaluated without writing a third-order parameter tensor. Differentiate the ordinary forward/backward program in the direction F .

Keep

$$u^\ell = D_F z^\ell, \quad g^\ell = D_F h^\ell = \phi'(z^\ell) u^\ell,$$

and define

$$\dot{\delta}^\ell := D_F \delta^\ell. \quad (6.12)$$

At the top hidden layer the exact readout velocity and exact directional boundary are

$$\dot{v}_r = -\frac{2}{B} \sum_c e_c h_r^3(c),$$

$$\dot{\delta}_r^3(a) = \dot{v}_r \phi'(z_r^3(a)) + v_r \phi''(z_r^3(a)) u_r^3(a). \quad (6.13)$$

Write a superscript (y) for the leading label-driven component of a directional state. Then

$$\dot{v}_r^{(y)} = \sum_c \lambda_c h_r^3(c),$$

$$\dot{\delta}_r^{3,(y)}(a) = \left(\sum_c \lambda_c h_r^3(c) \right) \phi'(z_r^3(a)) + v_r \phi''(z_r^3(a)) u_r^{3,(y)}(a). \quad (6.13y)$$

For $\ell = 1, 2$, recall $\delta^\ell = \phi'(z^\ell) \xi^\ell$. Hence

$$\dot{\delta}^\ell(a) = \phi''(z^\ell(a)) u^\ell(a) \xi^\ell(a) + \phi'(z^\ell(a)) \dot{\xi}^\ell(a). \quad (6.14)$$

Its label-driven component satisfies

$$\dot{\delta}^{\ell,(y)}(a) = \phi''(z^\ell(a)) u^{\ell,(y)}(a) \xi^\ell(a) + \phi'(z^\ell(a)) \dot{\xi}^{\ell,(y)}(a). \quad (6.14y)$$

The incoming signal is a transpose multiplication, so its exact finite-width directional derivative is

$$\dot{\xi}_j^\ell(a) = \frac{1}{\sqrt{n}} \sum_i \left[\dot{W}_{ij}^{\ell+1} \delta_i^{\ell+1}(a) + W_{ij}^{\ell+1} \dot{\delta}_i^{\ell+1}(a) \right]. \quad (6.15)$$

For the label-driven component,

$$\dot{W}_{ij}^{\ell+1,(y)} = \frac{1}{\sqrt{n}} \sum_c \lambda_c \delta_i^{\ell+1}(c) h_j^\ell(c).$$

Peeling its contribution to the first term of (6.15) uses the already-computed backward Gram:

$$\begin{aligned} \frac{1}{\sqrt{n}} \sum_i \dot{W}_{ij}^{\ell+1,(y)} \delta_i^{\ell+1}(a) &= \sum_c \lambda_c h_j^\ell(c) \frac{1}{n} \sum_i \delta_i^{\ell+1}(c) \delta_i^{\ell+1}(a) \\ &\rightarrow \sum_c \lambda_c h_j^\ell(c) \Pi_{ca}^{\ell+1}. \end{aligned} \quad (6.16)$$

The second term of (6.15) is a new backward transpose sum and must be peeled against every forward channel through the same matrix. Section 6.6 makes that operation explicit.

Now set

$$A := DF[F],$$

and define the correction-forward states

$$\bar{z}^\ell := D_A z^\ell, \quad \bar{h}^\ell := D_A h^\ell = \phi'(z^\ell) \bar{z}^\ell. \quad (6.17)$$

For concise denominators, let

$$\rho_1 = \sqrt{d_0}, \quad \rho_2 = \rho_3 = \sqrt{n}.$$

The exact differentiated vector field for a hidden matrix is

$$A_{W^\ell} = -\frac{2}{B\rho_\ell} \sum_c \left[\dot{f}_c \delta^\ell(c) h^{\ell-1}(c)^\top + e_c \delta^\ell(c) h^{\ell-1}(c)^\top + e_c \delta^\ell(c) g^{\ell-1}(c)^\top \right]. \quad (6.18)$$

Here $h^0(c) = x_c$ and $g^0 = 0$. In the first line of (6.18), the label-driven part of $\dot{f}_c \delta^\ell$ is odd in the readout and vanishes. Its initialization-output-driven part is even but belongs to the width-suppressed residual-change family identified below (6.8). Replacing e_c by $-y_c$ in the other two lines gives the leading even part

$$A_{W^\ell}^{\text{lead}} = \frac{1}{\rho_\ell} \sum_c \lambda_c \left[\delta^{\ell,(y)}(c) h^{\ell-1}(c)^\top + \delta^\ell(c) g^{\ell-1,(y)}(c)^\top \right]. \quad (6.19)$$

The exact correction-forward recursion is

$$\bar{z}^1(a) = A_{W^1} \frac{x_a}{\sqrt{d_0}}, \quad (6.20)$$

and, for $\ell = 2, 3$,

$$\bar{z}^\ell(a) = \frac{1}{\sqrt{n}} W^\ell \bar{h}^{\ell-1}(a) + \frac{1}{\sqrt{n}} A_{W^\ell} h^{\ell-1}(a). \quad (6.21)$$

Finally, the correction to the hidden Gram is

$$R_{n,ab}^\ell = \frac{1}{n} \sum_i \left[\bar{h}_i^\ell(a) h_i^\ell(b) + h_i^\ell(a) \bar{h}_i^\ell(b) \right], \quad (6.22a)$$

When $\bar{h}^{\ell,\text{lead}}$ is built from A^{lead} , the formal leading mean-field replacement is

$$R_{ab}^\ell = \mathbb{E} \left[\bar{h}^{\ell,\text{lead}}(a) h^\ell(b) + h^\ell(a) \bar{h}^{\ell,\text{lead}}(b) \right]. \quad (6.22)$$

Equations (6.13), (6.14), (6.15), (6.18), and (6.20)–(6.22a) are exact when they are built from the full directional states. Equations (6.13y), (6.16), (6.19), and the later mean-field recursions (6.29)–(6.35) describe only the leading label–label branch. Equation (6.22) additionally requires joint row convergence and uniform integrability. Together these identities turn the Hessian contraction into ordinary products of explicit forward, backward, and differentiated states.

6.5 The multichannel Wick–Stein rule required at step two

At one step, a matrix was coupled mainly to one forward and one backward channel. Differentiation creates several channels through the same matrix, so the single-channel shorthand (3.1) must be generalized without dropping any covariance.

Let W be an iid standard Gaussian $n \times n$ matrix. Suppose the same matrix creates transpose channels

$$b_j^\alpha = \frac{1}{\sqrt{n}} \sum_i W_{ij} y_i^\alpha, \quad \alpha = 1, \dots, q. \quad (6.23)$$

If the source vectors y^α and the remaining lower-group variables are frozen with respect to the current matrix W , and Ψ_j is a smooth function of all $b_j^1, \dots, b_j^q$, then the exact one-weight identity gives

$$\mathbb{E}[W_{ij} \Psi_j] = \frac{1}{\sqrt{n}} \sum_{\alpha=1}^q y_i^\alpha \mathbb{E}[\partial_{b^\alpha} \Psi_j]. \quad (6.24)$$

After summing j and exposing equality patterns, the mean-field forward multiplication has the form

$$\frac{1}{\sqrt{n}} \sum_j W_{ij} \Psi_j \implies \Gamma_\Psi + \sum_{\alpha=1}^q y_i^\alpha \mathbb{E}[\partial_{b^\alpha} \Psi]. \quad (6.25)$$

The row-distinct patterns form the centered Gaussian field Γ_Ψ . It is generated jointly with every other forward sum through W ; for two integrands Ψ and $\tilde{\Psi}$,

$$\mathbb{E}[\Gamma_\Psi(a)\Gamma_{\tilde{\Psi}}(b)] = \mathbb{E}[\Psi(a)\tilde{\Psi}(b)]. \quad (6.26)$$

All cross-covariances with previously created fields using the same matrix must also be retained. The transpose version is symmetric. If

$$z_i^\alpha = \frac{1}{\sqrt{n}} \sum_j W_{ij} x_j^\alpha,$$

and Υ_i depends on every z_i^α , then

$$\frac{1}{\sqrt{n}} \sum_i W_{ij} \Upsilon_i \implies \Xi_\Upsilon + \sum_{\alpha=1}^q x_j^\alpha \mathbb{E}[\partial_{z^\alpha} \Upsilon], \quad (6.27)$$

with joint fresh-field covariance

$$\mathbb{E}[\Xi_\Upsilon(a)\Xi_\Omega(b)] = \mathbb{E}[\Upsilon(a)\Omega(b)]. \quad (6.28)$$

Equations (6.25) and (6.27) are not independence substitutions. Their response terms are precisely the accumulated Stein/Onsager corrections; their fresh Gaussian terms are the surviving off-diagonal Wick families.

Equations (6.24)–(6.27) apply directly only when their channel sources are frozen with respect to the current matrix. If a source such as y^α or x^α itself depends on W , that dependence must first be exposed as additional forward or transpose channels, and the full integration-by-parts derivative must include it.

For a rigorous use of these rules, the list of channels must therefore be complete. A derivative such as $\partial_{z^\alpha} \Upsilon$ is a total derivative through all deterministic response pieces already built from that channel. If a fresh field made earlier with the same matrix reappears, it must be registered as another jointly generated Gaussian base channel rather than declared independent.

6.6 Peel the differentiated backward pass

Apply (6.27) to the label-driven component of the second term of (6.15). The visible forward channels through $W^{\ell+1}$ are the ordinary preactivation $z^{\ell+1}$, sourced by h^ℓ , and its first label tangent $u^{\ell+1,(y)}$, sourced by $g^{\ell,(y)}$. Thus the first two response terms are

$$\sum_c h_j^\ell(c) \mathbb{E} \left[\frac{\partial \delta^{\ell+1,(y)}(a)}{\partial z_c^{\ell+1}} \right]$$

and

$$\sum_c g_j^{\ell,(y)}(c) \mathbb{E} \left[\frac{\partial \delta^{\ell+1,(y)}(a)}{\partial u^{\ell+1,(y)}(c)} \right].$$

Together with the direct term (6.16), the peeled form is

$$\begin{aligned} \xi^{\ell,(y)}(a) = & \sum_c \lambda_c h^\ell(c) \Pi_{ca}^{\ell+1} + \Xi^{\ell,(y)}(a) \\ & + \sum_c h^\ell(c) \mathbb{E} \left[\partial_{z_c^{\ell+1}} \delta^{\ell+1,(y)}(a) \right] \\ & + \sum_c g^{\ell,(y)}(c) \mathbb{E} \left[\partial_{u^{\ell+1,(y)}(c)} \delta^{\ell+1,(y)}(a) \right] + \text{responses of any additional registered channel.} \end{aligned} \quad (6.29)$$

The Gaussian base field $\Xi^{\ell,(y)}$ is centered. It is generated jointly with the ordinary incoming field ξ^ℓ and all other transpose sums through the same matrix. In particular,

$$\mathbb{E}[\Xi^{\ell,(y)}(a)\Xi^{\ell,(y)}(b)] = \mathbb{E}[\delta^{\ell+1,(y)}(a)\delta^{\ell+1,(y)}(b)],$$

$$\mathbb{E}[\Xi^{\ell,(y)}(a)\xi^\ell(b)] = \mathbb{E}[\delta^{\ell+1,(y)}(a)\delta^{\ell+1}(b)]. \quad (6.30)$$

Equations (6.13y), (6.14y), and (6.29), evaluated in the order $3 \rightarrow 2 \rightarrow 1$, define the formal multichannel peel of the label-driven differentiated backward pass once a complete channel registry and covariance block are supplied. The placeholder in (6.29) makes this a schema, not a fully enumerated nonlinear closed recursion.

6.7 Peel the correction-forward pass

In the leading branch of (6.21), the term $W^\ell \bar{h}^{\ell-1,\text{lead}}/\sqrt{n}$ is a forward multiplication. The current matrix also generates the transpose channels $\xi^{\ell-1}$ and $\xi^{\ell-1,(y)}$, sourced respectively by δ^ℓ and $\delta^{\ell,(y)}$. Applying (6.25) gives

$$\begin{aligned}
\frac{1}{\sqrt{n}} W^\ell \bar{h}^{\ell-1, \text{lead}}(a) &\Rightarrow \bar{\Gamma}^{\ell, \text{lead}}(a) \\
&+ \sum_c \delta^\ell(c) \mathbb{E}[\partial_{\xi^{\ell-1}(c)} \bar{h}^{\ell-1, \text{lead}}(a)] \\
&+ \sum_c \delta^{\ell, (y)}(c) \mathbb{E}[\partial_{\xi^{\ell-1, (y)}(c)} \bar{h}^{\ell-1, \text{lead}}(a)] \\
&+ \text{responses of any additional registered channel.}
\end{aligned} \tag{6.31}$$

The fresh Gaussian base field $\bar{\Gamma}^{\ell, \text{lead}}$ must be generated jointly with $Z^\ell, u^{\ell, (y)}$, and every other forward sum through W^ℓ ; its covariance and all cross-covariances follow from (6.26).

The direct $A_{W^\ell} h^{\ell-1}$ term in (6.21) gives

$$\sum_c \lambda_c \delta^{\ell, (y)}(c) G_{ac}^{\ell-1} + \sum_c \lambda_c \delta^\ell(c) \frac{1}{n} \sum_j g_j^{\ell-1, (y)}(c) h_j^{\ell-1}(a). \tag{6.32}$$

The second empirical average is odd in the readout. Its disconnected limit is zero, and a connected two-copy peel is width-suppressed. Therefore the leading correction-forward recursion is

$$\bar{z}^{\ell, \text{lead}}(a) = \text{ForwardPeel}_\ell[\bar{h}^{\ell-1, \text{lead}}(a)] + \sum_c \lambda_c G_{ac}^{\ell-1} \delta^{\ell, (y)}(c), \tag{6.33}$$

where ForwardPeel means the right-hand side of (6.31) after a complete channel registry has been supplied, including every response and joint covariance. Since that registry is not fully enumerated here, (6.33) is a formal schema. For layer 1 there is no propagated term, so

$$\bar{z}^{1, \text{lead}}(a) = \sum_c \lambda_c G_{ac}^0 \delta^{1, (y)}(c). \tag{6.34}$$

With $\bar{h}^{\ell, \text{lead}} = \phi'(z^\ell) \bar{z}^{\ell, \text{lead}}$, substitution into (6.22) represents R^ℓ as fixed-dimensional Gaussian expectations. For example, once the differentiated-backward channel law has been supplied, the layer-1 endpoint is

$$\begin{aligned}
R_{ab}^1 = \sum_c \lambda_c \Big[&G_{ac}^0 \mathbb{E}[\phi(Z_b^1) \phi'(Z_a^1) \delta^{1, (y)}(c)] \\
&+ G_{bc}^0 \mathbb{E}[\phi(Z_a^1) \phi'(Z_b^1) \delta^{1, (y)}(c)] \Big].
\end{aligned} \tag{6.35}$$

The nonlinear variables inside (6.35) are explicit functions of the Gaussian base channels produced by the preceding forward, backward, first-tangent, and differentiated-backward peels. At fixed derivative order the Gaussian base-channel count is finite, so each layer requires an expectation over a fixed multiple qB of batch coordinates, independent of width. It need not be only one B -dimensional vector.

6.8 Execution schema for the general nonlinear correction

For clarity, the reusable two-step computation is the following finite-state program schema.

1. **Forward pass:** compute G^0, G^1, G^2, G^3 and retain representative Gaussian preactivations Z^1, Z^2, Z^3 .
2. **Ordinary backward peel:** compute ξ^3, δ^3 , then ξ^2, δ^2 , then ξ^1, δ^1 , retaining their complete joint covariances. This gives P^ℓ and Π^ℓ .
3. **First-tangent peel:** compute $u^{1, (y)}, g^{1, (y)}$, then $u^{2, (y)}, g^{2, (y)}$, then $u^{3, (y)}, g^{3, (y)}$ using the diagonal Stein responses and fresh off-diagonal Gaussian base fields in (5.39).
4. **Differentiated-backward peel:** start with the label-leading top boundary (6.13y). Then use (6.14y), (6.16), and the multichannel schema (6.29) for groups $3 \rightarrow 2 \rightarrow 1$.
5. **Correction-forward peel:** compute $\bar{z}^{1, \text{lead}}, \bar{h}^{1, \text{lead}}$ from (6.34), followed by layers 2 and 3 using (6.31)–(6.33). At each matrix, generate all fresh Gaussian base fields jointly with fields previously made from that matrix.
6. **Observable readout:** evaluate (6.22). Each entry is the expectation of an explicit function of the finite Gaussian base channels accumulated above.

The finite-width directional equations and the underlying multichannel one-weight integration-by-parts identity (6.24), under its frozen-source condition, are exact. The Gaussian-limit arrows (6.25) and (6.27) are formal mean-field conclusions under the assumptions in Section 2.7. The claim is that the Gaussian base channels converge jointly and every retained forward, backward, and differentiated state is an explicit nonlinear function of them—not that the entire state is jointly Gaussian. A theorem must additionally prove the leave-one-out estimates, uniform moment bounds, source-channel completeness, and joint convergence for the registered Gaussian base channels. The finite count q may depend on derivative order and on the chosen depth/state construction, but not on width. This qualification matters more at two steps than at one because differentiated fresh fields can be correlated with several earlier channels through the same matrix.

6.9 Two-step result

Subject to completion and convergence of the nonlinear channel registry in Section 6.8, the formal mean-field closure is, coefficient by coefficient,

$$\mathbb{E}[G_{n,ab}^{\ell, (2)}] = G_{ab}^\ell + \eta O(n^{-1}) + \eta^2 (4C_{ab}^\ell + R_{ab}^\ell) + o(\eta^2) + o_n(1). \tag{6.36}$$

For the second update alone,

$$\mathbb{E}[G_{n,ab}^{\ell,(2)} - G_{n,ab}^{\ell,(1)}] = \eta O(n^{-1}) + \eta^2(3C_{ab}^{\ell} + R_{ab}^{\ell}) + o(\eta^2) + o_n(1). \quad (6.37)$$

The expectations of both linear coefficients are $O(n^{-1})$ before the width limit; the random coefficients themselves need not be. The term C^{ℓ} is the one-step quadratic coefficient from Section 5, and R^{ℓ} is due entirely to recomputing the gradient. Equations (6.36)–(6.37) are formal for a general nonlinear activation and verified explicitly in the deep-linear specialization below.

6.10 Deep-linear audit of the two-step correction

Again set $\phi(z) = z$, let

$$\kappa = \frac{2}{B}, \quad \sigma = G^0 y,$$

and write

$$\tau(A) = \frac{1}{n} \operatorname{tr} A.$$

The one-step coefficients are

$$C_{ab}^{\ell} = (1, 5, 14)_{\ell} \kappa^2 \sigma_a \sigma_b.$$

For the correction term define

$$M_2 = \frac{1}{n} W^{2\top} W^2, \quad M_3 = \frac{1}{n} W^{3\top} W^3, \quad M_{23} = \frac{1}{n} W^{2\top} M_3 W^2. \quad (6.38)$$

Gaussian Wick contraction gives

$$\begin{aligned} \tau(M_2) &= \tau(M_3) = \tau(M_{23}) = 1, \\ \tau(M_2^2) &= \tau(M_3^2) = 2, \\ \tau(M_{23}M_2) &= 2, \quad \tau(M_{23}^2) = 3. \end{aligned} \quad (6.39)$$

For example, expanding

$$\tau(M_2^2) = \frac{1}{n^3} \sum_{a,b,i,j} W_{ia}^2 W_{ib}^2 W_{jb}^2 W_{ja}^2$$

produces three Wick pairings. Two retain three free indices and contribute 1 each; the crossing pairing retains only two and is $O(n^{-1})$. Hence $\tau(M_2^2) \rightarrow 2$.

Because M_{23} contains the independent Wishart matrix M_3 only once in $\tau(M_{23}M_2)$, conditioning on W^2 and using $\mathbb{E}[M_3] = I$ reduces it to $\tau(M_2^2)$, giving 2.

For $\tau(M_{23}^2)$, put $\tilde{M}_2 = W^2 W^{2\top} / n$. Cyclicity gives

$$\tau(M_{23}^2) = \tau(M_3 \tilde{M}_2 M_3 \tilde{M}_2).$$

For fixed symmetric $\tilde{M}_2$, the exact second moment of the Wishart entries is

$$\mathbb{E}[(M_3)_{ij}(M_3)_{kl}] = \delta_{ij}\delta_{kl} + \frac{1}{n}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}).$$

Substitution into the normalized trace yields, at leading order,

$$\mathbb{E}_{M_3}[\tau(M_3 \tilde{M}_2 M_3 \tilde{M}_2)] = \tau(\tilde{M}_2^2) + \tau(\tilde{M}_2)^2 + o(1) \rightarrow 2 + 1 = 3.$$

To connect these traces to the differentiated-state program, define the label-aggregated forward vectors

$$a^{\ell} := \sum_c \lambda_c h^{\ell}(c),$$

and the ordinary linear backward vectors

$$b^3 = v, \quad b^2 = \frac{1}{\sqrt{n}} W^{3\top} b^3, \quad b^1 = \frac{1}{\sqrt{n}} W^{2\top} b^2.$$

Because

$$a^2 = \frac{1}{\sqrt{n}} W^2 a^1, \quad a^3 = \frac{1}{\sqrt{n}} W^3 a^2,$$

differentiating the backward pass along the label-driven first update gives

$$\dot{b}^3 = a^3,$$

$$\dot{b}^2 = \frac{\|b^3\|^2}{n} a^2 + \frac{1}{\sqrt{n}} W^{3\top} a^3 \implies (I + M_3) a^2,$$

and

$$\dot{b}^1 = \frac{\|b^2\|^2}{n} a^1 + \frac{1}{\sqrt{n}} W^{2\top} b^2 \implies (I + M_2 + M_{23}) a^1. \quad (6.39a)$$

The first term in each line comes from differentiating the corresponding weight matrix; the remaining terms propagate the differentiated backward vector. Inserting (6.39a) into the correction-forward recursion (6.19)–(6.21) gives the following one-sided Gram contractions.

The one-sided correction contractions at layers 1, 2, and 3 are respectively

$$3 = \tau(I + M_2 + M_{23}),$$

$$7 = \tau[M_2(I + M_2 + M_{23})] + \tau(I + M_3) = (1 + 2 + 2) + (1 + 1),$$

and

$$\begin{aligned} 10 &= \tau[M_{23}(I + M_2 + M_{23})] + \tau[M_3(I + M_3)] + 1 \\ &= (1 + 2 + 3) + (1 + 2) + 1. \end{aligned}$$

The Gram has two endpoints, so symmetrization doubles these values:

$$\boxed{R_{ab}^1 = 6\kappa^2 \sigma_a \sigma_b, \quad R_{ab}^2 = 14\kappa^2 \sigma_a \sigma_b, \quad R_{ab}^3 = 20\kappa^2 \sigma_a \sigma_b.} \quad (6.40)$$

Therefore the cumulative two-step coefficients are

$$\boxed{(4C^\ell + R^\ell)_{ab} = (10, 34, 76)_\ell \kappa^2 \sigma_a \sigma_b,} \quad (6.41)$$

and the second-step increment coefficients are

$$\boxed{(3C^\ell + R^\ell)_{ab} = (9, 29, 62)_\ell \kappa^2 \sigma_a \sigma_b.} \quad (6.42)$$

This audit checks the factors 4 and 3, the gradient-correction sign, and the off-diagonal width patterns. It also distinguishes two Euler steps from gradient flow evaluated at time 2η : the latter would have quadratic coefficient $4C^\ell + 2R^\ell$.

7. What this case study changes in the peeling program

The three executions support the core peeling architecture, but they sharpen several statements that should be part of any theorem-level formulation.

1. **The raw/effective coordinate convention must be fixed before peeling.** In the raw convention used here,

$$\text{Cov}(W_{rp}^3, z_r^3(c) \mid h^2) = \frac{h_p^2(c)}{\sqrt{n}}.$$

Writing $h_p^2(c)/n$ would correspond to a different effective-weight convention and would corrupt width counting.

2. **Exact conditional Gaussian integration precedes deterministic Gram replacement.** A singled-out row may also occur in the empirical covariance of its layer. The safe proof uses leave-one-out or leave-two-out covariances, performs Wick–Stein first, and only then sends the empirical covariance to its deterministic limit.
3. **Stein-created lower-group factors remain boundary data.** For example, $h_p^2(c)h_q^2(d)$ created while peeling group 3 belongs to group 2. It cannot be averaged or summed out during the group-3 peel.
4. **An off-diagonal family cannot be removed from its one-copy mean.** In the backward kernel it is subleading after complete counting; in the quadratic feature coefficient it survives as the fresh field Γ^ℓ . The observable and the number of copies determine the answer.
5. **Expectation and concentration are separate proof obligations.** One-copy peeling identifies an annealed limit. Replacing empirical kernels inside later products requires a two-copy or higher-copy concentration argument plus uniform integrability.
6. **A limiting zero cannot be differentiated as zero.** Although $L_n^\ell \rightarrow 0$, its directional derivative contains the nonzero terms $2C^\ell + R^\ell$.
7. **Higher training steps require derivative-state augmentation.** At step two, the state must include $u, g, \dot{\delta}, \dot{\xi}, \bar{z}, \bar{h}$ in addition to the ordinary forward and backward variables. At higher Taylor order this becomes a finite jet of differentiated states.
8. **The Gaussian integration dimension is a fixed multiple of batch size.** A primitive forward kernel uses one B -dimensional Gaussian vector. A higher-order calculation uses a finite collection of jointly Gaussian base vectors, and the retained nonlinear states are explicit functions of them. Their total Gaussian dimension is qB . The number q may depend on derivative order and on the chosen depth/state construction, but not on width.
9. **An $O(L)$ recursion still requires depth-uniform finite-state closure.** The intended algorithm traverses depth a constant number of times at fixed jet order, but the nonlinear registry is still schematic and its base-channel count may depend on depth. Linear complexity in L is therefore a target conclusion, not yet an established result. It requires proving that the registered state size is bounded independently of depth at fixed derivative order.

10. **The nonlinear two-step closure still carries theorem obligations.** The exact finite-width directional identities are complete, but the displayed nonlinear mean-field channel registry is schematic. A rigorous theorem must enumerate the registry and prove joint convergence of its Gaussian base channels while tracking every cross-covariance; the deep-linear audit alone does not prove the general nonlinear closure.

8. Final consolidated statement

For this three-hidden-layer μP network, the mean-field peeling calculation gives the following coefficientwise results, subject to the concentration and joint-state obligations stated above:

$$\Pi^\ell = D^\ell \odot D^{\ell+1} \odot \dots \odot D^3,$$

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[\ell_a^+] = -\frac{4}{B} y_a \sum_c K_{ac} y_c,$$

$$\lim_{n \rightarrow \infty} [\eta] \mathbb{E}[G_{n,ab}^{\ell,+}] = 0,$$

$$\lim_{n \rightarrow \infty} [\eta^2] \mathbb{E}[G_{n,ab}^{\ell,+}] = C_{ab}^\ell,$$

and

$$\lim_{n \rightarrow \infty} [\eta^2] \mathbb{E}[G_{n,ab}^{\ell,(2)}] = 4C_{ab}^\ell + R_{ab}^\ell.$$

The backward kernels and one-step coefficient C^ℓ close through explicit fixed-batch Gaussian expectations under the stated joint-CLT assumptions. The two-step correction R^ℓ is represented by the exact differentiated-state identities and formally reduced through the multichannel Gaussian-peeling schema in Section 6. Its general nonlinear registry and convergence remain proof obligations, while its full deep-linear specialization is explicitly verified in (6.40)–(6.42).

Part IV. Theorem and Literature Positioning

Mean-Field Peeling, Tensor Programs, and the Sharp Theorem Claim

Status: Research-positioning report

Date: 2026-08-03

Scope: Contrast with Tensor Programs, defensible novelty framing, sharp target theorem, and naming recommendations

Epistemic status: The literature comparison is an evidence-based assessment, not a certification of novelty. The proposed theorem below is a target statement and must not be presented as proved until its observable grammar, closure, termination, limit justification, and complexity bounds have been established.

1. Executive conclusion

Tensor Programs does not merely prove that a suitable infinite-width limit exists. It gives constructive recursive semantics for a finite tensor computation: coordinatewise operations are transported to limiting coordinate variables, normalized width averages become expectations, matrix multiplication introduces jointly Gaussian source variables with recursively computed covariances, and reuse of a matrix and its transpose produces an explicit Onsager correction.

Consequently, the following is not a defensible novelty claim for mean-field peeling:

Tensor Programs only proves that the answer exists, whereas mean-field peeling is the first method that shows how to compute it recursively as Gaussian expectations.

Tensor Programs already shows how at the level of recursive limiting semantics.

The meaningful distinction is instead between two levels of construction:

1. **Recursive limiting semantics:** associate recursively defined limiting random variables and scalar expectations to a finite tensor computation.
2. **Explicit Gaussian normal form:** eliminate all wide random objects, recursive limit variables, unresolved Onsager terms, neuron-index sums, and hidden dependence corrections, leaving a finite directed acyclic graph of fully specified Gaussian integrals.

Mean-field peeling can constitute a distinct contribution if it proves a source-to-source compilation theorem of the second kind for a precisely defined class of contracted higher-derivative observables. Its potential contribution is therefore not a new infinite-width existence principle. It is an explicit derivative-observable elimination calculus, a Gaussian normal-form theorem, and possibly a finite-state depth recursion with controlled complexity.

The recommended paper title is:

Mean-Field Peeling: Explicit Gaussian Normal Forms for Feature-Learning Jets

The recommended method name is **mean-field derivative peeling**, shortened in prose to **mean-field peeling**. The recommended theorem name is **Mean-Field Derivative Peeling Theorem**. The output representation should be called the **Gaussian normal form**, and the finite collection of local Taylor coefficients should be called the **feature-learning jet**.

2. Claim ladder

The relevant claims should be separated rather than blended into one headline.

- **Normalized wide-network averages can have deterministic infinite-width limits.** Established broadly in prior work; not the proposed novelty.
- **A finite tensor computation can be assigned recursive limiting coordinate semantics and moment expectations.** Established by Tensor Programs; not the proposed novelty.
- **Discrete-time feature-learning limits in μP can be represented and computed through Tensor Programs.** Established in substance by Tensor Programs IV; not the proposed novelty.

- **First- and higher-order derivative kernels admit recursive analyses.** Significant prior art exists; not by itself the proposed novelty.
- **Every observable in a specified contracted derivative grammar can be mechanically reduced to an explicit Gaussian-integral normal form.** Open target claim; plausible contribution.
- **The reduction terminates by eliminating one maximal layer group at a time.** Open general theorem; supported by worked examples only.
- **For fixed batch size, derivative order, training-step count, and observable schema, the normal form closes on finitely many state types independent of width and depth.** Open and decisive claim.
- **After common-state sharing, the number of recursion nodes is at most a fixed constant times depth.** Open and potentially important algorithmic contribution.
- **Every Gaussian atom is exactly B -dimensional for a batch of size B .** Not currently established; should not be claimed without a factorization theorem.
- **Fixed Taylor coefficients determine positive-time feature-learning dynamics.** False without a separate convergence or analyticity theorem; outside the peeling theorem.

3. What “computable recursively” means in Tensor Programs

Tensor Programs supplies mathematically specified operational rules. In simplified notation, if a coordinatewise operation creates

$$x_i = \phi(y_i^1, \dots, y_i^k),$$

then its limiting coordinate variable is defined recursively by

$$Z^x = \phi(Z^{y^1}, \dots, Z^{y^k}).$$

If a scalar is a normalized width average,

$$q_n = \frac{1}{n} \sum_{i=1}^n \psi(x_i^1, \dots, x_i^k),$$

then its limiting scalar is defined by

$$q_n \longrightarrow \mathbb{E}[\psi(Z^{x^1}, \dots, Z^{x^k})].$$

For a Gaussian matrix W reused in a computation, Tensor Programs decomposes the limiting coordinate of Wx into a fresh Gaussian component and an Onsager component,

$$Z^{Wx} = \widehat{Z}^{Wx} + \dot{Z}^{Wx}.$$

The fresh components associated with the same matrix are jointly Gaussian, with covariance of the schematic form

$$\text{Cov}(\widehat{Z}^{Wx}, \widehat{Z}^{Wy}) = \sigma_W^2 \mathbb{E}[Z^x Z^y].$$

When $W^\top$ has occurred earlier in the program, the correction is defined through expected symbolic derivatives. Schematically,

$$\dot{Z}^{Wx} = \sigma_W^2 \sum_a Z^{y_a} \mathbb{E} \left[\frac{\partial Z^x}{\partial \widehat{Z}^{W^\top y_a}} \right].$$

This is a constructive Stein/Onsager rule. It is not an assertion that an unspecified correction exists.

Algorithm 1 in the Tensor Programs IV supplement instructs the reader to write training and inference as a Tensor Program and recursively compute the limiting coordinate variable associated with every vector and the limiting value associated with every scalar. The same supplement explicitly discusses the symbolic unwinding needed to calculate the Onsager term and notes that the calculation can grow rapidly in complexity.

Thus Tensor Programs occupies the following level of explicitness:

Given an admissible finite program, recursive rules define its limiting coordinate law and all scalar moments in that program.

This is stronger than an existence theorem.

4. What remains implicit or unnormalized in Tensor Programs

Although the semantics is constructive, several nontrivial compilation tasks can remain.

4.1 The derivative observable must first be represented

Tensor Programs begins with a straight-line computation involving vectors, matrices, scalars, coordinatewise functions, and moments. A mathematical expression containing an arbitrary contracted Hessian or higher parameter derivative is not automatically a Tensor Program input.

One must first:

- differentiate or expand the network computation;
- encode every derivative contraction in the permitted program language;
- determine the required width normalization;
- verify that the resulting computation satisfies the hypotheses of the Master Theorem.

Closure of a general higher-derivative observable grammar under this compilation is not itself the generic Tensor Programs theorem.

4.2 Limiting coordinate variables may remain recursively named

An answer of the form

$$\mathbb{E}[\Psi(Z^1, \dots, Z^m)]$$

need not yet be a Gaussian normal form. The variables $Z^1, \dots, Z^m$ can be nonlinear functions of earlier Gaussian sources and Onsager terms. To obtain an ordinary Gaussian integral, one must unwind them as

$$(Z^1, \dots, Z^m) = F(G), \quad G \sim N(0, \Sigma),$$

and then write

$$\mathbb{E}[\Psi(Z^1, \dots, Z^m)] = \int \Psi(F(g)) N(0, \Sigma)(dg).$$

This unrolling is possible for a fixed finite program in the denotational sense, but Tensor Programs generally permits the recursive Z representation to remain as the answer.

4.3 The symbolic Onsager calculation can be burdensome

The correction $\dot{Z}^{W^x}$ requires expressing Z^x in terms of the underlying Gaussian source variables, differentiating that expression, and taking another expectation. The Tensor Programs IV supplement explicitly notes that this calculation can balloon quickly and that the general terms have no easy expression.

This is not a defect in mathematical definition. It is the distinction between a well-defined recursive interpreter and a compact, normalized symbolic output.

4.4 “Finite-dimensional” is not automatically “batch-dimensional”

For Tensor Programs, a fixed finite computation introduces finitely many Gaussian source variables. Their number can nevertheless grow with depth, training time, number of forward and backward passes, or derivative order.

Therefore the statement

the answer is a finite-dimensional Gaussian expectation

does not by itself imply that every integral is over only one B -dimensional preactivation vector, nor that the dimension is uniformly bounded in depth.

5. Intended contribution of mean-field peeling

Mean-field peeling starts from a different input and demands a more normalized output.

5.1 Input

The proposed input is a scalar contracted derivative observable constructed from network outputs, losses, hidden feature Grams, preactivations, activations, and a fixed number of parameter or training-trajectory derivatives.

5.2 Transformation

The proposed compiler performs, explicitly:

1. derivative expansion by chain and product rules;
2. recursive scalarization of the current maximal layer;
3. grouping by layer;
4. conditional Gaussian integration over the maximal-layer weights;
5. Stein expansion of every weight–preactivation dependence;
6. Wick contraction of the remaining Gaussian weights;
7. enumeration of neuron-index equality partitions;
8. width-power counting and removal of strictly subleading partitions;
9. replacement of surviving row averages by their deterministic mean-field Gaussian moments;
10. propagation of the resulting lower-layer boundary state to the next peel.

5.3 Output

The desired output is a finite directed acyclic graph containing only deterministic algebra and Gaussian atoms of the form

$$\mathcal{J}_{\ell, F} = \int_{\mathbb{R}^d} F(g) N(0, K^\ell)(dg),$$

where all of the following are explicitly generated:

- the layer ℓ ;
- the covariance matrix K^ℓ ;
- the dimension d ;
- the integrand F ;
- every occurrence of $\phi, \phi', \dots$;
- every deterministic coefficient;
- every surviving batch-index contraction and equality pattern.

The output should contain no random weights, neuron-index sums, implicit backward vectors, recursively named Tensor Program variables, or unevaluated Onsager terms.

On their common domain, the two calculations should agree. The intended relationship is

$$\text{Peel}(O) = \text{UnrollTP}(\text{CompileDerivative}(O)).$$

Mean-field peeling can therefore be viewed as a specialized normal-form compiler or layerwise variable-elimination calculus for derivative observables, not as a competing infinite-width semantics.

6. Recommended sharp theorem contract

The theorem should first be stated for fully connected MLPs with equal hidden width. Extensions to unequal widths, residual networks, convolutions, attention, normalization, or general Tensor Programs should be separate results.

6.1 Network and limit

Fix:

- a batch of B input-label pairs;

- a depth L ;
- a derivative order r ;
- a fixed number of gradient steps when training updates are included;
- an MLP with hidden width n and fully declared μP initialization and optimizer scaling;
- a scalar activation with sufficiently many polynomially bounded derivatives.

Take $n \rightarrow \infty$ with B, L , derivative order, training-step count, and the observable schema fixed. A complexity statement may subsequently compare the symbolic construction across values of L , but the probability limit itself is width first at every fixed depth.

6.2 Admissible derivative observables

An admissible derivative observable should be defined structurally. It is a scalar expression generated from:

- outputs and losses on the fixed batch;
- preactivations and activations;
- hidden-layer Gram entries;
- parameter derivatives of total order at most r ;
- finite sums, products, and tensor contractions;

subject to:

1. all neuron and parameter indices are eventually contracted;
2. every width normalization is declared;
3. the local expression schema is independent of width;
4. depth dependence is generated by repeated layer-local templates;
5. the expression is width-balanced under the declared scaling.

The phrase “any observable having a mean-field limit” should not define the class, because that would make the theorem partly circular.

6.3 Target theorem

Target theorem: Mean-Field Derivative Peeling.

For every admissible scalar derivative observable O_n , there is a deterministic syntactic transformation Peel that recursively eliminates the maximal remaining layer group. Each peel replaces that group by a finite sum of deterministic Gaussian-integral coefficients multiplying admissible lower-group states. The transformation terminates after at most L layer eliminations and emits a finite Gaussian normal-form DAG. Under the stated regularity and uniform-integrability hypotheses,

$$\lim_{n \rightarrow \infty} \mathbb{E}_{\theta_0}[O_n] = \text{Eval}(\text{Peel}(O_n)).$$

Every integrand, covariance, coefficient, index partition, and activation derivative in the output is effectively constructed from the network specification, fixed data, and observable syntax.

This is the conservative core claim. It asserts a finite explicit construction but does not yet assert that its size is linear in depth.

6.4 Finite-state strengthening

The strongest useful extension is:

For every fixed batch size, derivative order, training-step count, and observable schema, the lower-layer boundary signatures produced by peeling close on a finite set of state types. After identical states are shared, the normal-form DAG has at most

$$C_{B,r,O} L$$

nodes, and every Gaussian atom has dimension at most

$$d_{B,r,O},$$

where the effective constants are independent of width and depth.

The constants may grow combinatorially or exponentially with derivative order and training-step count. No polynomial dependence should be claimed without a separate analysis.

This finite-state statement is the sharpest contrast with generic Tensor Program unrolling, but it is also the most important open proof obligation. Until it is proved, the safe statement is “at most L elimination stages,” not “an $O(L)$ -size formula.”

6.5 Possible batch-dimensional strengthening

A still stronger corollary would assert that every connected Gaussian atom can be reduced to one B -dimensional layer-preactivation integral, so that all multiple-row contributions factor into products of such atoms.

That statement may be true for the intended initialization derivative grammar, but it requires a factorization theorem for the equality partitions created by Wick and Stein expansions. The current program should use the safe bound $d_{B,r,O}$ until this is proved.

7. Feature-learning-jet corollary

Consider full-batch gradient descent with canonical μP scaling and master learning rate η . Fix a number s of gradient steps and a Taylor order p .

For sample a , let $\mathcal{L}_a^{(s)}(\eta)$ be its loss after s steps. For layer ℓ , define the hidden feature Gram

$$G_{ab}^{\ell,(s)}(\eta) = \frac{1}{n} \left\langle h_a^{\ell,(s)}(\eta), h_b^{\ell,(s)}(\eta) \right\rangle.$$

For each $k \leq p$, the coefficients

$$[\eta^k] \mathbb{E}[\mathcal{L}_a^{(s)}(\eta)]$$

and

$$[\eta^k] \mathbb{E}[G_{ab}^{\ell,(s)}(\eta)]$$

are initialization derivative observables of fixed order. Once their admissibility is proved, the peeling theorem gives explicit Gaussian normal forms for them.

The finite collection of coefficients through order p is the **order- p feature-learning jet**.

This corollary makes a precise claim about every fixed coefficient. It does not imply convergence of the Taylor series, a positive radius of convergence, or correctness of a truncated series at a fixed nonzero learning rate.

8. Required nonclaims

The theorem and abstract should state explicitly that the work does not prove:

- convergence of the learning-rate or time Taylor series;
- complete positive-time training dynamics;
- arbitrary or width-dependent training times;
- a uniform-in-depth probability limit;
- elementary closed forms for general activation functions;
- polynomial complexity in derivative order or training-step count;
- self-averaging of every admissible observable;
- general architectural universality;
- exactly B -dimensional integration without the additional factorization proof.

Expectation-level exactness and sample-level determinism must remain separate claims. A variance or concentration theorem is required to pass from the former to the latter.

9. Reviewer-safe contrast with Tensor Programs

The recommended contrast is:

- **Input.** Tensor Programs takes a straight-line tensor computation; mean-field derivative peeling takes a contracted derivative observable.
- **Output semantics.** Tensor Programs gives recursive limiting coordinate semantics; peeling gives a completely eliminated Gaussian-integral normal form.
- **Intermediate randomness.** Tensor Programs introduces recursively defined Z variables; peeling removes all random intermediate variables from the final representation.
- **Onsager correction.** Tensor Programs defines it through expected symbolic derivatives; peeling executes the Stein branches explicitly during layer elimination.
- **Dimension claim.** In Tensor Programs, “finite-dimensional” means finite for a fixed finite program; peeling seeks a bound depending only on fixed batch/order/schema data.
- **Compression claim.** Generic Tensor Program unrolling does not promise a compact normal form; peeling seeks finite-state closure and a depth-linear shared DAG.

The recommended prose statement is:

Tensor Programs supplies general recursive limiting semantics for finite neural computations. Mean-field derivative peeling is a normal-form compiler for contracted derivative observables: it eliminates the wide random computation one layer at a time and emits a finite DAG of explicitly specified Gaussian integrals.

The work should not claim that Tensor Programs is nonconstructive. The distinction is between a recursive semantic representation and a normalized symbolic elimination theorem.

10. Naming and branding recommendations

10.1 Recommended vocabulary

- **Method:** mean-field derivative peeling
- **Short method name:** mean-field peeling
- **Main theorem:** Mean-Field Derivative Peeling Theorem
- **Output representation:** Gaussian normal form
- **Finite local Taylor data:** feature-learning jet
- **Algorithmic structure:** peeling-state DAG or peeling dynamic program

The acronym **MFP** should be avoided because it is already naturally read as “mean-field parameterization.” If an acronym is needed, **MFDP** is less ambiguous, but the full phrase is preferable in exposition.

10.2 Recommended paper title

Mean-Field Peeling: Explicit Gaussian Normal Forms for Feature-Learning Jets

This is the preferred title because it combines:

- the memorable method name;
- the concrete output claim;
- the local, fixed-order nature of the feature-learning application.

10.3 Alternative titles

1. **Mean-Field Derivative Peeling: An Explicit Gaussian Calculus for Local Feature Learning**
2. **Peeling Wide Neural Networks: Explicit Gaussian Recursions for Feature-Learning Derivatives**
3. **From Derivatives to Gaussian Integrals: Mean-Field Peeling in Feature-Learning Networks**
4. **Feature-Learning Jets at Infinite Width: A Mean-Field Peeling Calculus**

If “jet” is considered too unfamiliar for the target venue, the safest replacement is “Taylor coefficients”:

Mean-Field Peeling: Explicit Gaussian Normal Forms for Feature-Learning Taylor Coefficients

10.4 Alternative theorem names

1. **Mean-Field Derivative Peeling Theorem** — recommended canonical name.
2. **Gaussian Normal-Form Peeling Theorem** — best descriptive name for the output result.
3. **Layerwise Gaussian Elimination Theorem for Derivative Observables** — precise but less memorable.
4. **Feature-Learning Jet Representation Theorem** — best reserved for the training corollary, not the core initialization calculus.

11. Recommended paper framing

The paper should be framed as a constructive refinement and specialization, not as a replacement for Tensor Programs.

A reviewer-resistant abstract-level framing is:

Existing infinite-width frameworks provide recursive limiting semantics for broad classes of neural computations. We study a narrower but algebraically difficult class: scalar contractions involving fixed-order derivatives of deep mean-field networks. We introduce mean-field derivative peeling, a layerwise conditional Gaussian elimination calculus that compiles such observables into explicit Gaussian normal forms. The procedure executes all Stein corrections, Wick contractions, equality-partition bookkeeping, and width power counting, leaving no wide random objects or implicit limit variables. For fixed batch size and derivative order, we identify the finite peeling states required by local feature-learning coefficients and derive explicit recursions for their Gaussian-integral representations.

The final sentence should only claim finite-state or depth-linear recursions after those properties have been proved.

12. Decisive proof obligations

The highest-leverage unresolved obligations are:

1. **Observable grammar:** define the admissible contracted derivative language without referring circularly to the existence of its limit.
2. **Derivative closure:** prove that chain/product-rule expansion and every peel remain inside a finite extension of that language.
3. **Stein termination:** prove that repeated dependence removal terminates and produces only the declared activation derivatives and lower-group states.
4. **Equality-partition completeness:** prove that all leading neuron-index patterns are enumerated and no apparently subleading pattern can regain order through a lower-layer contraction.
5. **Mean-field replacement:** justify every empirical-average-to-expectation step with the required conditional law of large numbers, uniform integrability, or concentration estimate.
6. **Finite-state closure:** classify the lower-layer boundary signatures at fixed derivative order.
7. **Complexity:** prove the shared-DAG bound and state exactly how its constant depends on derivative order, batch size, training steps, and observable schema.
8. **Gaussian dimension:** determine whether the safe finite bound can be strengthened to one B -dimensional atom per connected row component.
9. **Tensor Programs equivalence:** on the overlap class, prove that peeling evaluates to the same limit as the Tensor Program semantics. This converts a possible reviewer objection into a validation theorem.

The single most decisive next step is the finite-state closure theorem. Without it, peeling is an explicit layerwise elimination recipe but not yet the claimed $O(L)$ dynamic program. With it, the method has a precise and substantial distinction from generic symbolic unrolling.

13. Primary literature anchors

- Greg Yang and Edward J. Hu, Tensor Programs IV: Feature Learning in Infinite-Width Neural Networks¹⁸, ICML 2021.
- Greg Yang and Edward J. Hu, Tensor Programs IV supplementary material¹⁹, especially Definition G.3, the Master Theorem, Algorithm 1, and the discussion of computational complexity.
- Greg Yang, Tensor Programs II: Neural Tangent Kernel for Any Architecture²⁰, 2020.
- Greg Yang, Tensor Programs III: Neural Matrix Laws²¹, 2020.
- Jiaoyang Huang and Horng-Tzer Yau, Dynamics of Deep Neural Networks and Neural Tangent Hierarchy²², ICML 2020.
- Ethan Dyer and Guy Gur-Ari, Asymptotics of Wide Networks from Feynman Diagrams²³, 2019.
- Max Guillen, Philipp Misof, and Jan E. Gerken, Finite-Width Neural Tangent Kernels from Feynman Diagrams²⁴, ICML 2026.

14. Current authoritative positioning statement

The strongest defensible current statement is:

Mean-field peeling is a proposed explicit normal-form calculus for fixed-order contracted derivative observables in wide mean-field MLPs. Tensor Programs already supplies constructive recursive semantics for broad finite neural computations, including μP feature-learning limits. The prospective distinction is that peeling begins from derivative syntax and performs all layerwise Gaussian elimination, Stein correction, Wick contraction, equality-pattern enumeration, and width power counting explicitly, with the goal of producing a finite Gaussian-integral DAG and a finite-state depth recursion. The existence of this normal form is an open target theorem; the depth-linear finite-state strengthening remains the decisive unproved claim.

¹⁸<https://proceedings.mlr.press/v139/yang21c.html>

¹⁹<https://proceedings.mlr.press/v139/yang21c/yang21c-supp.pdf>

²⁰<https://arxiv.org/abs/2006.14548>

²¹<https://arxiv.org/abs/2009.10685>

²²<https://proceedings.mlr.press/v119/huang201.html>

²³<https://arxiv.org/abs/1909.11304>

²⁴<https://arxiv.org/abs/2508.11522>